IB CHEMISTRY HL

Curated Past-Paper Review

137 questions (87 multiple choice, 50 structured), picked from the 1067 in the full package as the hardest and/or highest-quality ones. Questions reused across exam sessions appear only once.
Labels (e.g. 1.2-S17) are unchanged, so they still match the full package and its markscheme.
Answers are in the companion Curated Markscheme PDF, in the same order.
A copy of the IB Chemistry data booklet is needed for many questions.

★ top priority (74 questions): the hardest / best ones - do these first if short on time

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Notes

1.1 Inorganic Nomenclature is not included: all 16 of its questions are one-step recall.

9.3-S1: the source markscheme cuts off the first method for (f)(i); the second method is complete.

LESSON 1.2

Stoichiometry

5 multiple-choice · 9 structured

Part A - Multiple choice

★ **1.2-M7** May 2024 TZ2 · Paper 1 · Q3

3. 6.00 mol of copper, Cu, are mixed with 12.00 mol of dilute nitric acid, $HNO_3(aq)$. The equation for the reaction that occurs is shown below.

$$3Cu(s) + 8HNO_3(aq) \rightarrow 3Cu(NO_3)_2(aq) + 2NO(g) + 4H_2O(l)$$

What is the amount, in mol, of nitrogen(II) oxide, NO, produced assuming that the reaction goes to completion?

A. 3.00

B. 4.00

C. 8.00

D. 18.00

1.2-M9 Nov 2023 TZ1 · Paper 1 · Q2

2. Metal M reacts with 16.0 g of sulfur to produce 26.0 g of the compound MS_2. What is the relative atomic mass of M?

A. 5

B. 10

C. 20

D. 40

1.2-M10 Nov 2023 TZ1 · Paper 1 · Q3

3. 64 g of methane and 96 g of oxygen are reacted according to the equation.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

What would be found in the reaction vessel at completion of the reaction?

A. $CO_2(g)$ and $H_2O(l)$ only

B. $O_2(g)$, $CO_2(g)$ and $H_2O(l)$ only

C. $CH_4(g)$, $CO_2(g)$ and $H_2O(l)$ only

D. $CH_4(g)$, $O_2(g)$, $CO_2(g)$ and $H_2O(l)$

1.2-M43 May 2015 TZ1 · Paper 1 · Q3

3. Which solution contains the biggest amount, in mol, of chloride ions?

A. $20\,cm^3$ of $0.50\,mol\,dm^{-3}\,NH_4Cl$

B. $60\,cm^3$ of $0.20\,mol\,dm^{-3}\,MgCl_2$

C. $70\,cm^3$ of $0.30\,mol\,dm^{-3}\,NaCl$

D. $100\,cm^3$ of $0.30\,mol\,dm^{-3}\,ClCH_2COOH$

1.2-M60 May 2012 TZ2 · Paper 1 · Q2

2. Nitroglycerine, $C_3H_5N_3O_9$, can be used in the manufacture of explosives. What is the coefficient of $C_3H_5N_3O_9(l)$ when the equation for its decomposition reaction is balanced using the lowest whole numbers?

$$___\,C_3H_5N_3O_9(l) \rightarrow ___\,CO_2(g) + ___\,H_2O(l) + ___\,N_2(g) + ___\,O_2(g)$$

A. 2

B. 4

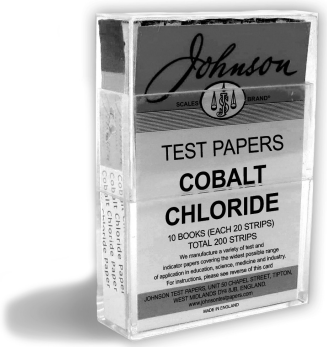
C. 20

D. 33

Part B - Structured questions

1.2-S6 May 2025 TZ3 · Paper 1B · Q3 parts (b)(i), (b)(v)

3. Cobalt(II) chloride has been used for many years as a qualitative test for water, H_2O.



Hydrated cobalt(II) chloride has the formula $CoCl_2 \cdot \mathbf{X}H_2O$ where **X** can be 1, 2, 6 or 9.

A teacher set a task to determine the value of **X**. Hydrated cobalt(II) chloride is pink in colour and anhydrous cobalt(II) chloride is blue.

A group of students weighed 10.000 g of hydrated cobalt(II) chloride and left it overnight in an oven at 30 °C. They observed a slight colour change from the original pink as a result.

Each student heated the hydrated cobalt(II) chloride until no further observable change was seen, allowed it to cool and recorded the mass.

The following equation illustrates the process occurring:

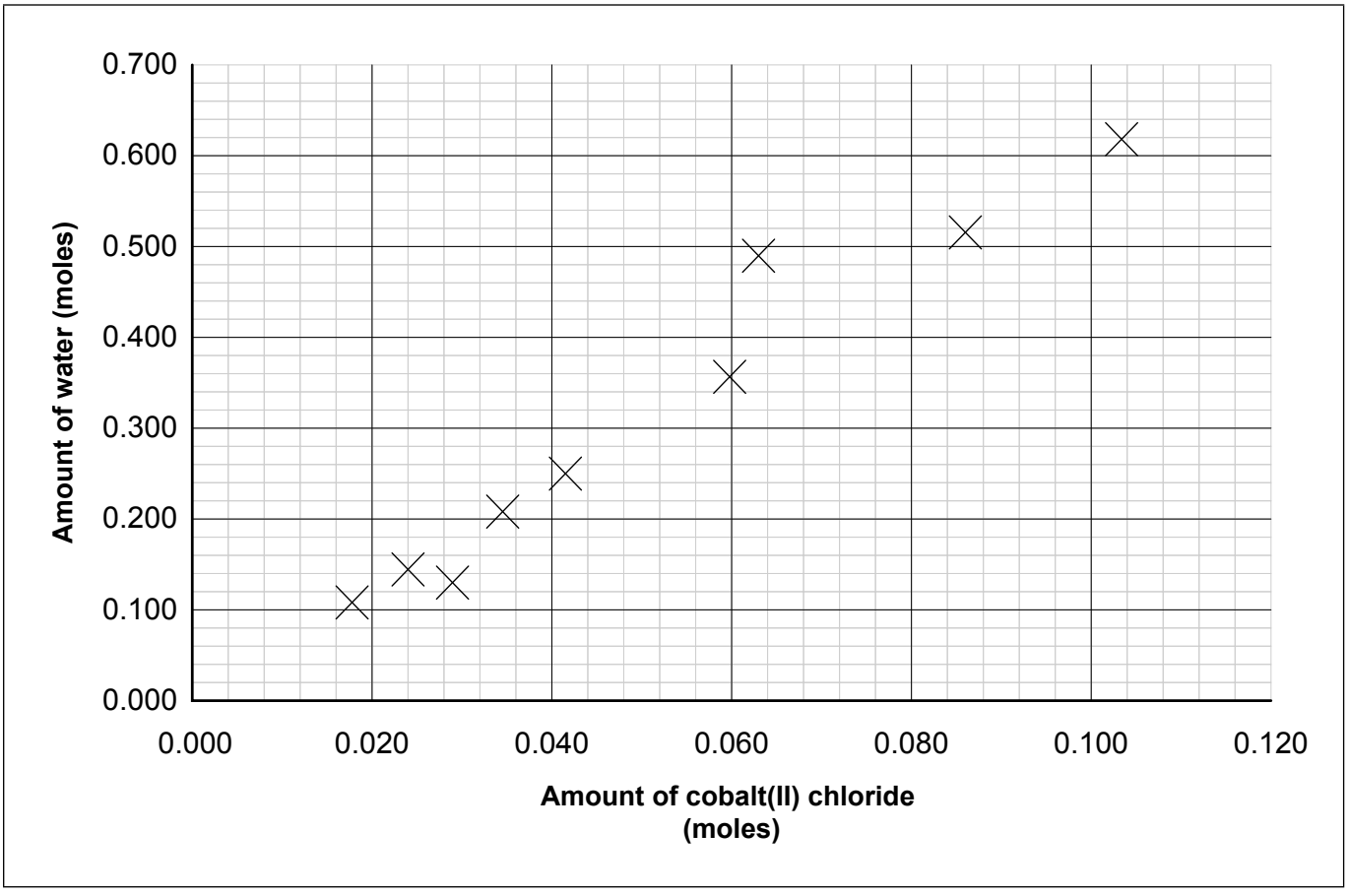
$$CoCl_2 \cdot \mathbf{X}H_2O(s) \rightarrow CoCl_2(s) + \mathbf{X}H_2O(g)$$

(b) Each of the students carried out the same experiment using different masses of hydrated cobalt(II) chloride. The shared results are shown below.

Experiment number	Mass of $CoCl_2 \cdot XH_2O$	Mass of $CoCl_2$	Loss in mass (g)	Amount of $CoCl_2$ / mol	Amount of H_2O / mol
1	4.256	2.299	1.957	0.018	0.109
2	5.711	3.117	2.594	0.024	0.144
3	6.853	4.501	2.352	0.029	0.131
4	8.224	4.470	3.754	0.035	0.208
5	9.868	5.372	4.496	0.041	0.250
6	11.842	6.481		0.050	
7	14.210	7.761	6.449	0.060	0.358
8	17.052	8.222	8.830	0.063	0.490
9	20.463	11.172	9.291	0.086	0.516
10	24.555	13.396	11.159	0.103	0.619

(i) Determine the missing values in the table above (M_r $H_2O = 18.02\,g\,mol^{-1}$). [1]

A graph of the results was plotted.



(v) Determine the value of **X**. [1]

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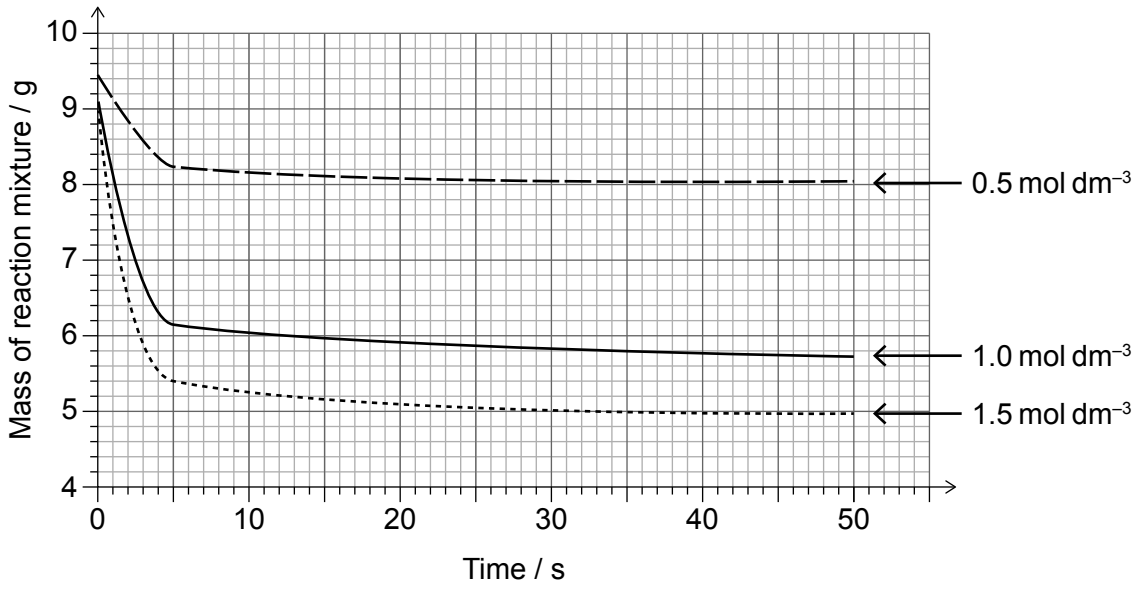
1.2-S3 May 2025 TZ2 · Paper 1B · Q3 parts (b)

3. The reaction of solid sodium hydrogencarbonate, $NaHCO_3(s)$, with hydrochloric acid, $HCl(aq)$, was investigated using the same mass of solid and a constant volume of different concentrations of acid. A data logger was used to measure the mass of the reaction mixture every five seconds.

The results were graphed.



(b) Suggest, with a reason, whether $NaHCO_3(s)$ **or** $HCl(aq)$ was the limiting reagent in the reactions using 0.5 and 1.0 mol dm^{-3} $HCl(aq)$. [1]

Limiting reagent: . . . . . .

Reason: . . . . . .

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1.2-S17 May 2022 TZ1 · Paper 2 · Q1 parts (a)(i), (b)(i), (b)(iii), (c)(i), (c)(ii), (d)(i), (d)(iii), (e)(iv) · also 2.1, 6.1, 7.3

1. When heated in air, magnesium ribbon reacts with oxygen to form magnesium oxide.

(a) (i) Write a balanced equation for the reaction that occurs. [1]

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(b) The reaction in (a)(i) was carried out in a crucible with a lid and the following data was recorded:

Mass of crucible and lid = 47.372 ±0.001 g

Mass of crucible, lid and magnesium ribbon before heating = 53.726 ±0.001 g

Mass of crucible, lid and product after heating = 56.941 ±0.001 g

(i) Calculate the amount of magnesium, in mol, that was used. [1]

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(ii) Suggest an explanation, other than product being lost from the crucible or reacting with nitrogen, that could explain the yield found in (b)(iii). [1]

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(d) The presence of magnesium nitride can be demonstrated by adding water to the product. It is hydrolysed to form magnesium hydroxide and ammonia.

(i) Calculate coefficients that balance the equation for the following reaction. [1]

$$__\ Mg_3N_2(s) + __\ H_2O(l) \rightarrow __\ Mg(OH)_2(s) + __\ NH_3(aq)$$

(iii) Determine the oxidation state of nitrogen in Mg_3N_2 and in NH_3. [1]

Mg_3N_2: . . . . . .

NH_3: . . . . . .

(e) Most nitride ions are $^{14}N^{3-}$.

(iv) Suggest, giving a reason, whether magnesium or nitrogen would have the greater sixth ionization energy. [1]

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★ **1.2-S21** Nov 2019 · Paper 2 · Q2 parts (a)(i), (a)(ii), (a)(iii)

2. The biochemical oxygen demand of a water sample can be determined by the following series of reactions. The final step is titration of the sample with sodium thiosulfate solution, $Na_2S_2O_3(aq)$.

$$2Mn^{2+}(aq) + O_2(aq) + 4OH^-(aq) \rightarrow 2MnO_2(s) + 2H_2O(l)$$

$$MnO_2(s) + 2I^-(aq) + 4H^+(aq) \rightarrow Mn^{2+}(aq) + I_2(aq) + 2H_2O(l)$$

$$2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$

A student analysed two 300.0 cm^3 samples of water taken from the school pond: one immediately (day 0), and the other after leaving it sealed in a dark cupboard for five days (day 5). The following results were obtained for the titration of the samples with 0.0100 $mol\,dm^{-3}$ $Na_2S_2O_3(aq)$.

Sample	Titre / cm^3 $\pm 0.1\,cm^3$
Day 0	25.8
Day 5	20.1

(a) (i) Determine the mole ratio of $S_2O_3^{2-}$ to O_2, using the balanced equations. [1]

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(ii) Calculate the number of moles of oxygen in the day 0 sample. [2]

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(iii) The day 5 sample contained 5.03×10^{-5} moles of oxygen.

Determine the 5-day biochemical oxygen demand of the pond, in mg dm^{-3} ("parts per million", ppm). [2]

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1.2-S26 Nov 2018 · Paper 2 · Q1 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (d) · also 7.2, 8.2

1. 3.26 g of iron powder are added to 80.0 cm^3 of 0.200 mol dm^{-3} copper(II) sulfate solution. The following reaction occurs:

$$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$$

(a) (i) Determine the limiting reactant showing your working. [2]

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(ii) The mass of copper obtained experimentally was 0.872 g. Calculate the percentage yield of copper. [2]

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(b) (i) The reaction was carried out in a calorimeter. The maximum temperature rise of the solution was 7.5 °C.

Calculate the enthalpy change, ΔH, of the reaction, in kJ, assuming that all the heat released was absorbed by the solution. Use sections 1 and 2 of the data booklet. [2]

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(ii) State another assumption you made in (b)(i). [1]

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(d) A student electrolyzed aqueous iron(II) sulfate, $FeSO_4(aq)$, using platinum electrodes. State half-equations for the reactions at the electrodes, using section 24 of the data booklet. [2]

Anode (positive electrode):

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Cathode (negative electrode):

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1.2-S35 Nov 2016 · Paper 2 · Q2 parts (c)(ii), (c)(iii), (c)(iv), (d), (e) · also 3.1, 4.1

2. The concentration of a solution of a weak acid, such as ethanedioic acid, can be determined by titration with a standard solution of sodium hydroxide, NaOH (aq).

(c) 5.00 g of an impure sample of hydrated ethanedioic acid, $(COOH)_2 \bullet 2H_2O$, was dissolved in water to make 1.00 dm^3 of solution. 25.0 cm^3 samples of this solution were titrated against a 0.100 mol dm^{-3} solution of sodium hydroxide using a suitable indicator.

$$(COOH)_2(aq) + 2NaOH(aq) \rightarrow (COONa)_2(aq) + 2H_2O(l)$$

The mean value of the titre was 14.0 cm^3.

(ii) Calculate the amount, in mol, of NaOH in 14.0 cm^3 of 0.100 $mol\,dm^{-3}$ solution. [1]

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(iii) Calculate the amount, in mol, of ethanedioic acid in each 25.0 cm^3 sample. [1]

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(iv) Determine the percentage purity of the hydrated ethanedioic acid sample. [3]

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(d) Draw the Lewis (electron dot) structure of the ethanedioate ion, $^{-}OOCCOO^{-}$. [1]

(e) Outline why all the C–O bond lengths in the ethanedioate ion are the same length and suggest a value for them. Use section 10 of the data booklet. [2]

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1.2-S38 May 2016 · Paper 2 · Q1 parts (a)(i), (a)(ii), (a)(iv), (a)(v), (b)(i), (b)(iii), (b)(iv), (c)(i), (c)(ii), (c)(iii), (d)(i) · also 3.1, 4.2, 5.1, 5.2, 6.1, 8.2

1. Phosphine (IUPAC name phosphane) is a hydride of phosphorus, with the formula PH_3.

(a) (i) Draw a Lewis (electron dot) structure of phosphine. [1]

(ii) State the hybridization of the phosphorus atom in phosphine. [1]

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(iv) Outline whether you expect the bonds in phosphine to be polar or non-polar, giving a brief reason. [1]

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(v) Phosphine has a much greater molar mass than ammonia. Explain why phosphine has a significantly lower boiling point than ammonia. [2]

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(b) Phosphine is usually prepared by heating white phosphorus, one of the allotropes of phosphorus, with concentrated aqueous sodium hydroxide. The equation for the reaction is:

$$P_4(s) + 3OH^-(aq) + 3H_2O(l) \rightarrow PH_3(g) + 3H_2PO_2^-(aq)$$

(i) The first reagent is written as P_4, not 4P. Describe the difference between P_4 and 4P. [1]

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(iii) State the oxidation state of phosphorus in P_4 and $H_2PO_2^-$. [2]

P_4:

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$H_2PO_2^-$:

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(iv) Oxidation is now defined in terms of change of oxidation number. Explore how earlier definitions of oxidation and reduction may have led to conflicting answers for the conversion of P_4 to $H_2PO_2^-$ and the way in which the use of oxidation numbers has resolved this. [3]

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(c) 2.478 g of white phosphorus was used to make phosphine according to the equation:

$$P_4(s) + 3OH^-(aq) + 3H_2O(l) \rightarrow PH_3(g) + 3H_2PO_2^-(aq)$$

(i) Calculate the amount, in mol, of white phosphorus used. [1]

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(ii) This phosphorus was reacted with 100.0 cm^3 of 5.00 mol dm^{-3} aqueous sodium hydroxide. Deduce, showing your working, which was the limiting reagent. [1]

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(iii) Determine the excess amount, in mol, of the other reagent. [1]

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(d) Impurities cause phosphine to ignite spontaneously in air to form an oxide of phosphorus and water.

(i) 200.0 g of air was heated by the energy from the complete combustion of 1.00 mol phosphine. Calculate the temperature rise using section 1 of the data booklet and the data below. [1]

Standard enthalpy of combustion of phosphine, $\Delta H_c^\ominus = -750\,kJ\,mol^{-1}$

Specific heat capacity of air $= 1.00\,J\,g^{-1}K^{-1} = 1.00\,kJ\,kg^{-1}K^{-1}$

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★ **1.2-S43** May 2014 TZ2 · Paper 2 · Q1 parts (a), (c)(ii), (c)(iii), (g) · also 5.2

1. A class studied the equilibrium established when ethanoic acid and ethanol react together in the presence of a strong acid, using propanone as an inert solvent. The equation is given below.

$$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$$

One group made the following **initial mixture**:

Liquid	Volume / cm^3
Ethanoic acid	5.00 ± 0.05
Ethanol	5.00 ± 0.05
$6.00\,mol\,dm^{-3}$ aqueous hydrochloric acid	1.00 ± 0.02
Propanone	39.0 ± 0.5

(a) The density of ethanoic acid is $1.05\,g\,cm^{-3}$. Determine the amount, in mol, of ethanoic acid present in the initial mixture. *[3]*

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(c) After one week, a $5.00 \pm 0.05\,cm^3$ sample of the final equilibrium mixture was pipetted out and titrated with $0.200\,mol\,dm^{-3}$ aqueous sodium hydroxide to determine the amount of ethanoic acid remaining. The following titration results were obtained:

Titration number	1	2	3
Initial reading / cm^3 ± 0.05	1.20	0.60	14.60
Final reading / cm^3 ± 0.05	28.80	26.50	40.70
Titre / cm^3	27.60	25.90	26.10

(ii) Suggest the average volume of alkali, required to neutralize the $5.00\,cm^3$ sample, that the student should use. *[1]*

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(iii) $3.00\,cm^3$ of the $0.200\,mol\,dm^{-3}$ aqueous sodium hydroxide reacted with the hydrochloric acid present in the $5.00\,cm^3$ sample. Determine the concentration of ethanoic acid in the final equilibrium mixture. *[2]*

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(g) Propanone is used as the solvent because one compound involved in the equilibrium is insoluble in water. Identify this compound and explain why it is insoluble in water. *[2]*

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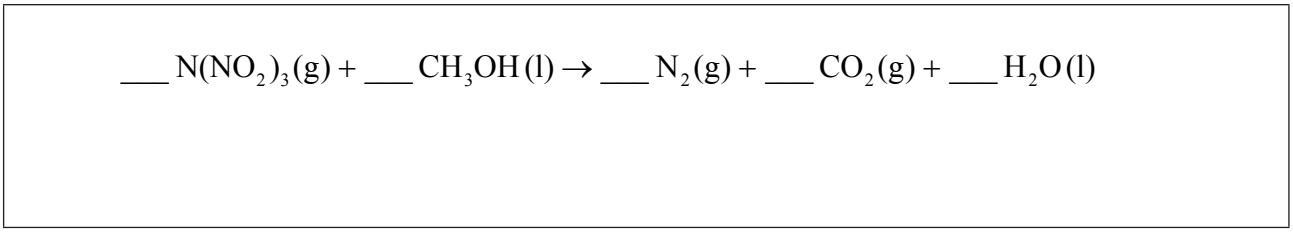
1.2-S44 Nov 2013 · Paper 2 · Q5 parts (a), (c), (g), (h), (i) · also 3.1, 4.1, 5.1, 9.3

5. In December 2010, researchers in Sweden announced the synthesis of N,N–dinitronitramide, $N(NO_2)_3$. They speculated that this compound, more commonly called trinitramide, may have significant potential as an environmentally friendly rocket fuel oxidant.

(a) Methanol reacts with trinitramide to form nitrogen, carbon dioxide and water. Deduce the coefficients required to balance the equation for this reaction. *[1]*



(c) Calculate the enthalpy change, in $kJ\,mol^{-1}$, when one mole of trinitramide decomposes to its elements, using bond enthalpy data from Table 10 of the Data Booklet. Assume that all the N–O bonds in this molecule have a bond enthalpy of $305\,kJ\,mol^{-1}$. *[3]*

(g) Outline how the length of the N–N bond in trinitramide compares with the N–N bond in nitrogen gas, N_2. *[2]*

(h) Deduce the N–N–N bond angle in trinitramide and explain your reasoning. *[3]*

(i) Predict, with an explanation, the polarity of the trinitramide molecule. *[2]*

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LESSON 2.1

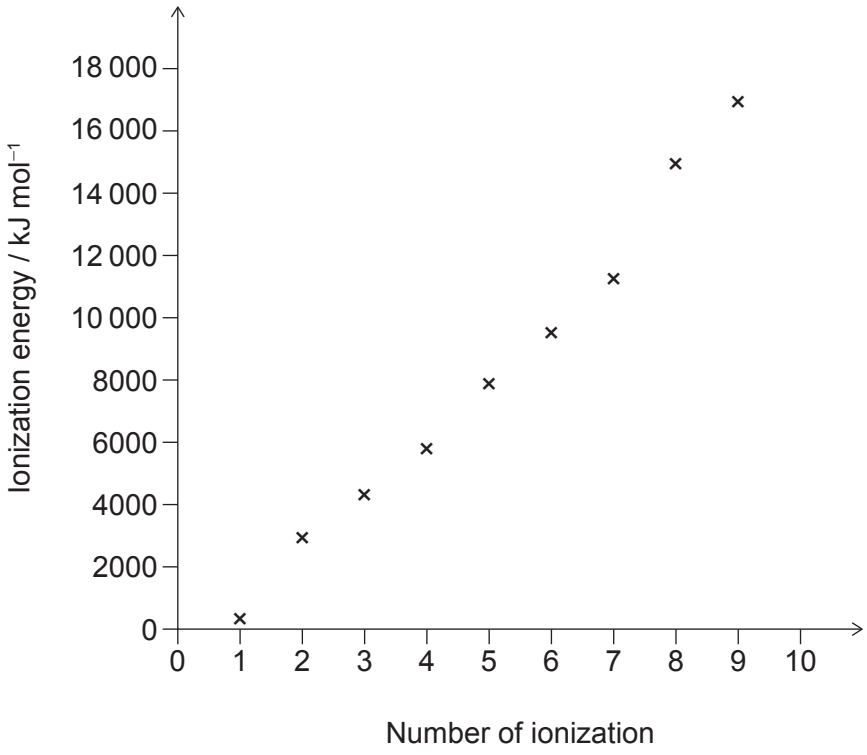
Ionization Energy

11 multiple-choice · 4 structured

Part A - Multiple choice

2.1-M5 May 2025 TZ2 · Paper 1A · Q4

4. What is the group number of the element whose successive ionization energies are shown?



A. 1

B. 2

C. 13

D. 14

2.1-M19 Nov 2022 · Paper 1 · Q4

4. Successive ionization energies of an element, **X**, are shown.

	1st	2nd	3rd	4th
Ionization energy (kJ mol^{-1})	740	1450	7730	10 540

What energy, in kJ mol^{-1}, is required for element **X** to reach its most stable oxidation state in ionic compounds?

A. 740

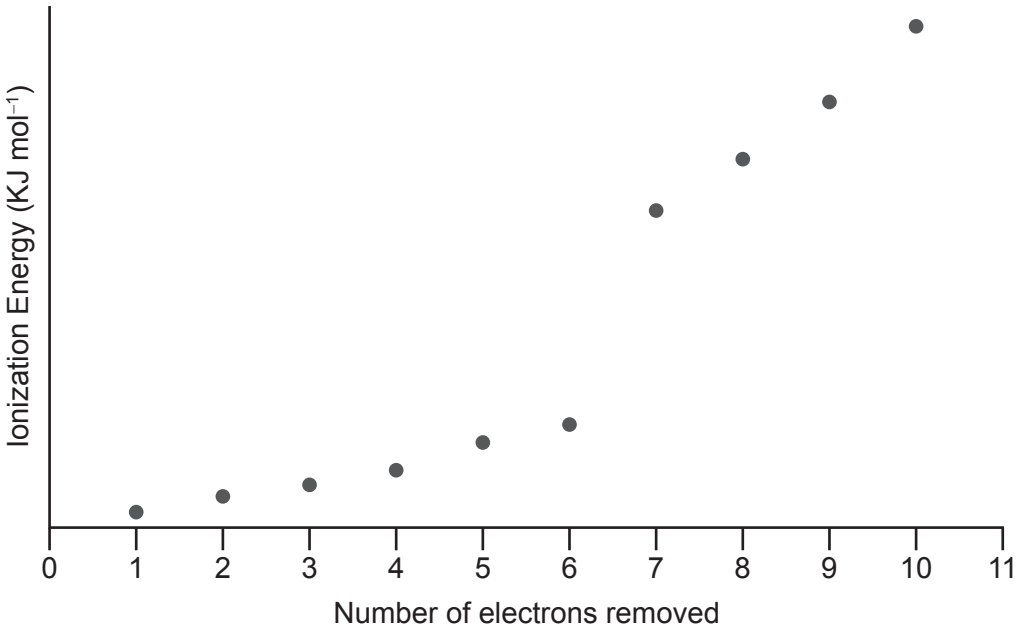
B. 1450

C. 2190

D. 7730

2.1-M10 Nov 2024 · Paper 1 · Q7

7. The graph represents the first ten ionization energies (IE) of an element.



What is the element?

A. Cl

B. Ne

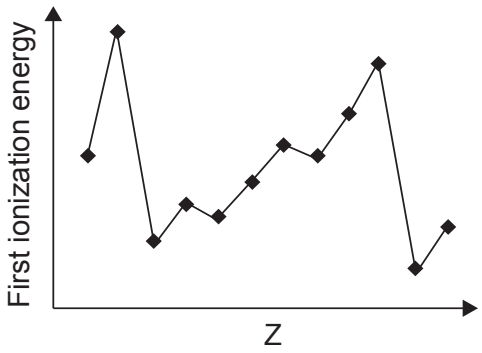
C. O

D. S

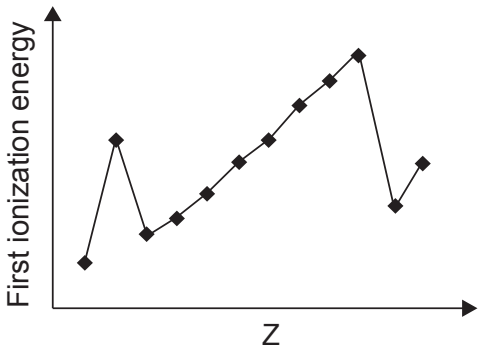
2.1-M13 May 2024 TZ2 · Paper 1 · Q6

6. Which graph shows the correct trend in the first ionization energies of the first twelve elements in the periodic table?

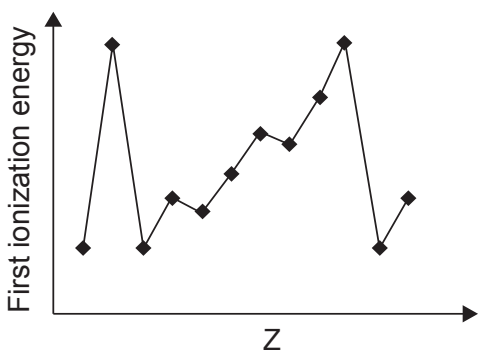
A.



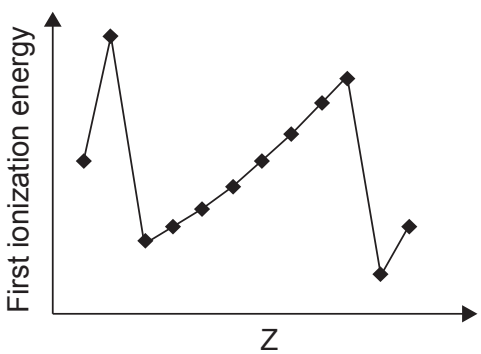
B.



C.

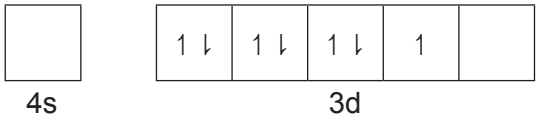


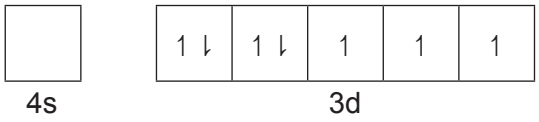
D.

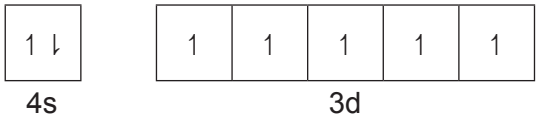


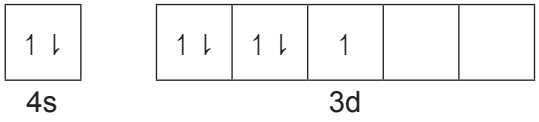
2.1-M14 May 2024 TZ2 · Paper 1 · Q9

9. In which diagram are the 4s and 3d orbitals of the Co^{2+} ion correctly filled with electrons?

A. 

B. 

C. 

D. 

2.1-M22 Nov 2021 · Paper 1 · Q5

5. Which statement explains why the **second** ionization energy of aluminium is higher than the **first** ionization energy of magnesium?

A. Ionization energy increases along period 3.

B. 3p electrons are at a higher energy level than 3s electrons.

C. 3p electrons are further away from the nucleus than 2p electrons.

D. Both have the same number of electrons and aluminium has one more proton.

★ **2.1-M32** Nov 2018 · Paper 1 · Q7

7. Which species will require the least energy for the removal of one electron?

A. Na^+

B. Mg^+

C. Al^{2+}

D. C^{3+}

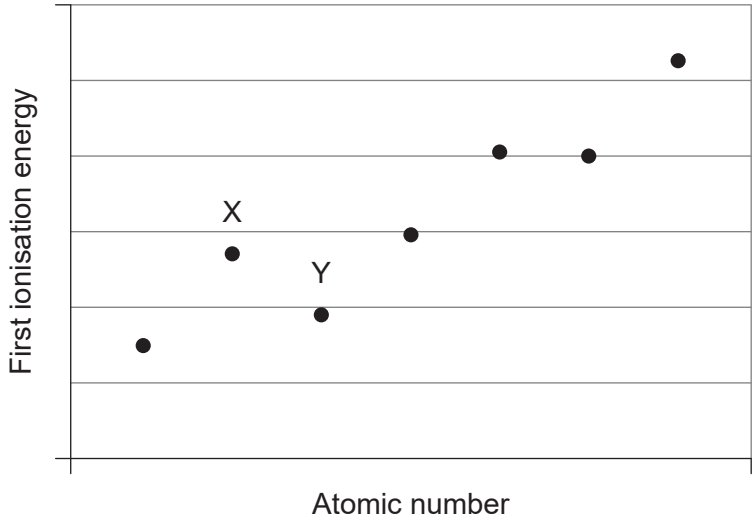
2.1-M57 May 2012 TZ1 · Paper 1 · Q8

8. What are the electron configurations of Cu, Cu^+ and Cu^{2+}?

	Cu	**Cu^+**	**Cu^{2+}**
A.	$[Ar]\ 4s^2\ 3d^9$	$[Ar]\ 4s^2\ 3d^8$	$[Ar]\ 4s^2\ 3d^7$
B.	$[Ar]\ 4s^2\ 3d^9$	$[Ar]\ 4s^1\ 3d^9$	$[Ar]\ 3d^9$
C.	$[Ar]\ 4s^2\ 3d^9$	$[Ar]\ 3d^{10}$	$[Ar]\ 3d^9$
D.	$[Ar]\ 4s^1\ 3d^{10}$	$[Ar]\ 3d^{10}$	$[Ar]\ 3d^9$

2.1-M23 May 2021 TZ1 · Paper 1 · Q6

6. The diagram shows the first ionisation energies of consecutive elements in the same period of the periodic table.



Which factor explains why element X has a higher first ionisation energy than element Y?

A. Element Y loses an electron from a different sub-level.

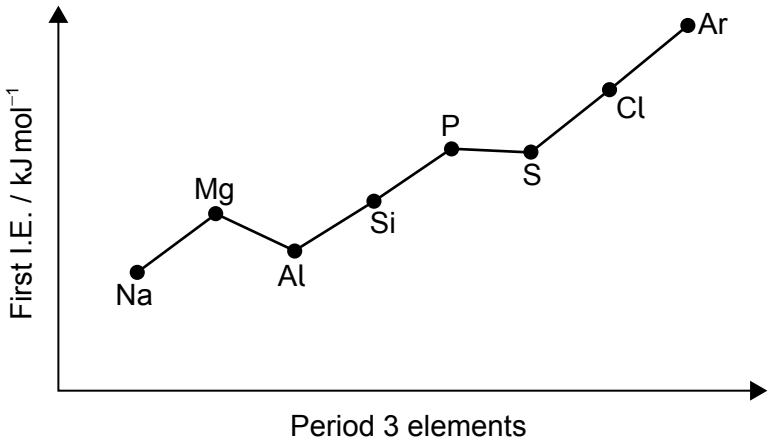
B. Element Y has a smaller atomic radius.

C. Element X has a full octet.

D. Element Y has a greater nuclear charge.

2.1-M38 May 2017 TZ1 · Paper 1 · Q5

5. Which statement explains one of the decreases in first ionization energy (I.E.) across period 3?



A. The nuclear charge of element Al is greater than element Mg.

B. The electron-electron repulsion is greater, for the electron with the opposite spin, in element S than in element P.

C. A new sub-level is being filled at element S.

D. The p orbital being filled in element Al is at a lower energy than the s orbital in element Mg.

2.1-M51 May 2014 TZ1 · Paper 1 · Q5

5. Which electron configurations do **not** follow the Hund's rule?

	1s	2s	2p		
I.	↑ ↓	↑ ↓	↑	↑	↑
II.	↑ ↓	↑ ↓	↑ ↓	↑	
III.	↑ ↓	↑ ↓	↑	↓	↑

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

Part B - Structured questions

★ **2.1-S3** May 2025 TZ2 · Paper 2 · Q2 parts (b)(iii), (b)(vii), (b)(viii), (c)(i), (c)(ii), (c)(iii) · also 6.3, 7.2, 9.4

2. Thallium is a heavy metal in group 13 of the periodic table.

(b) Thallium (I) sulfate has the formula Tl_2SO_4.

(iii) State the enthalpy term that characterises the strength of the bonding between the ions in an ionic solid. [1]

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Anode (positive electrode): . . . . . .

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Cathode (negative electrode): . . . . . .

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(c) Thallium has an atomic emission spectrum with a strong green line.

(i) State the feature of the atomic emission spectrum that may be used to find the ionization energy of the element. [1]

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(ii) Calculate the wavelength that corresponds to the ionization energy of thallium. Use sections 1, 2 and 9 of the data booklet. [3]

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(iii) Explain why, in terms of nuclear charge and the shielding of the valence electrons, the first ionization energy of thallium is lower than that of lead. [2]

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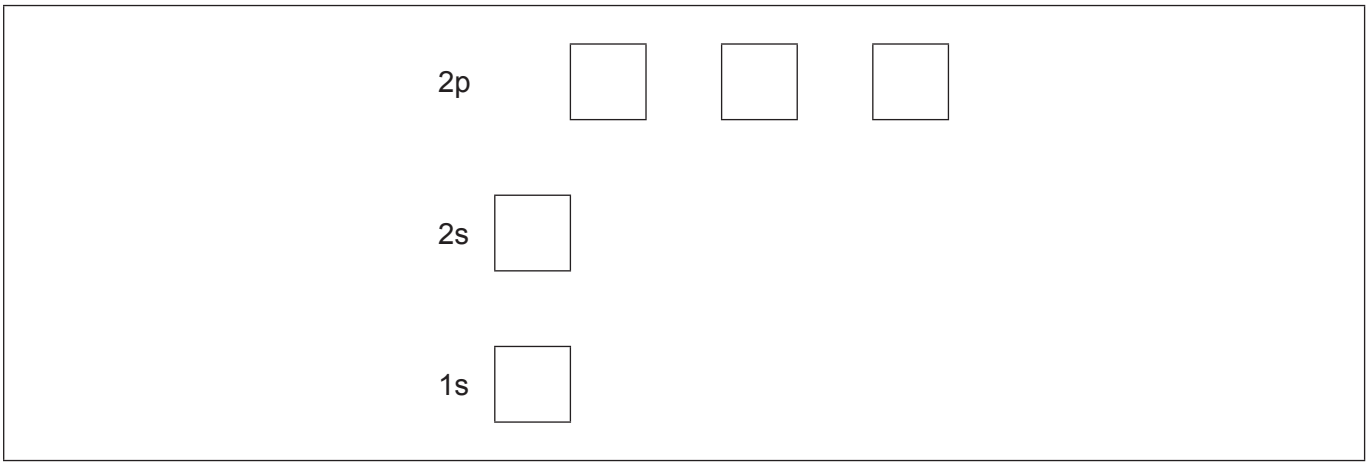
2.1-S7 May 2022 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(ii), (a)(iii) · also 3.1, 4.1

6. Nitric acid is usually produced by the oxidation of ammonia.

(a) (i) Draw arrows in the boxes to represent the electron configuration of a nitrogen atom. [1]



(ii) Deduce a Lewis (electron dot) structure of the nitric acid molecule, HNO_3, that obeys the octet rule, showing any non-zero formal charges on the atoms. [2]

(iii) Explain the relative lengths of the three bonds between N and O in nitric acid. [3]

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2.1-S13 May 2021 TZ2 · Paper 2 · Q2 parts (a)(ii), (b)(i), (c), (d) · also 3.1, 5.2

2. The properties of elements can be predicted from their position in the periodic table.

(ii) Explain why the first ionization energy of sulfur is lower than that of phosphorus. [2]

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(b) (i) State the condensed electron configurations for Cr and Cr^{3+}. [2]

Cr:

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Cr^{3+}:

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(c) Deduce the Lewis (electron dot) structure and molecular geometry of sulfur tetrafluoride, SF_4, and sulfur dichloride,SCl_2. [4]

Species	SF_4	SCl_2
Lewis structure		
Molecular geometry	. . . . .	. . . . .

(d) Suggest, giving reasons, the relative volatilities of SCl_2 and H_2O. [3]

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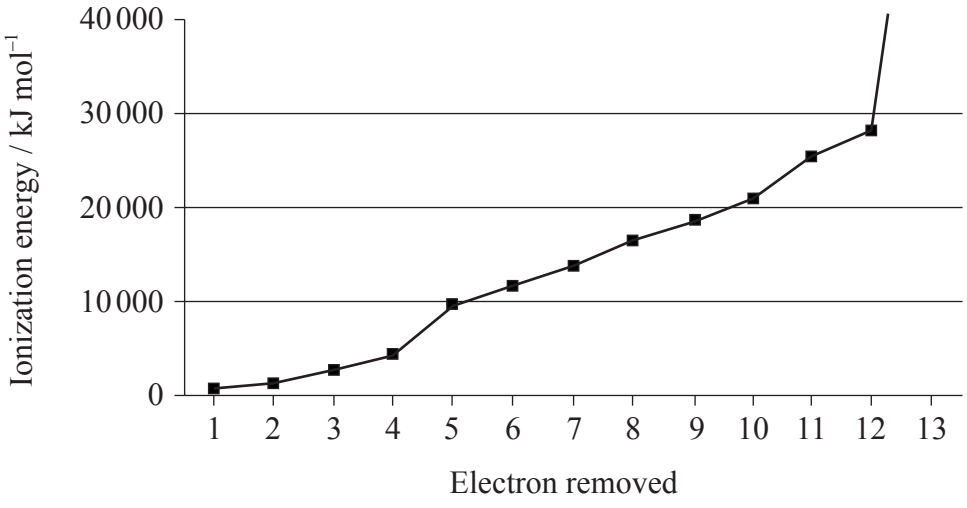
2.1-S31 May 2012 TZ1 · Paper 2 · Q8 parts (b)(i), (b)(ii)

8. The element titanium is present in meteorites.

(b) The successive ionization energies of titanium are shown below.



(i) State the **full** electron configuration of an atom of titanium and identify the sub-level from which the electron is removed when the 1st ionization energy is measured. *[2]*

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(ii) Explain why there are relatively large differences between the 4th and 5th, **and** between the 10th and 11th ionization energies. *[3]*

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LESSON 3.1

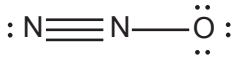
VSEPR Theory

7 multiple-choice · 3 structured

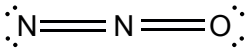
Part A - Multiple choice

3.1-M2 May 2025 TZ1 · Paper 1A · Q10

10. Nitrogen(I) oxide, N_2O, can be represented by two Lewis formulas.

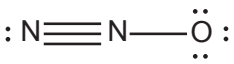
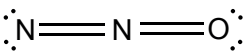


Structure X



Structure Y

Which row correctly identifies the formal charges and the more stable structure?

	Structure X 			Structure Y 			More stable
A.	0	+1	−1	−1	+1	0	X
B.	−1	+1	0	+1	0	−1	X
C.	0	+1	−1	−1	0	+1	Y
D.	−1	0	+1	0	+1	−1	Y

3.1-M6 May 2024 TZ1 · Paper 1 · Q11

11. Which combination best describes NO_2^-?

	Number of electron domains around N	Molecular geometry	Bond angle
A.	2	linear	180°
B.	3	bent	105°
C.	3	bent	117°
D.	4	trigonal planar	120°

3.1-M16 Nov 2022 · Paper 1 · Q11

11. Which molecule has a tetrahedral molecular geometry?

A. HNO_3

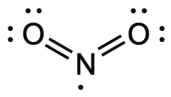
B. SF_4

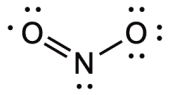
C. XeF_4

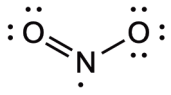
D. XeO_4

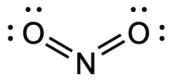
3.1-M14 May 2023 TZ2 · Paper 1 · Q12

12. Which is the valid and preferred Lewis structure of the NO_2• radical, based on formal charge?

A. 

B. 

C. 

D. 

3.1-M24 Nov 2018 · Paper 1 · Q9

9. Which species has the same molecular geometry as SO_3^{2-}?

A. BF_3

B. SO_3

C. PF_3

D. CO_3^{2-}

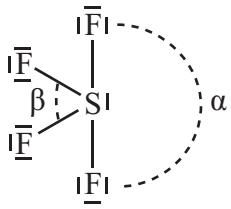
3.1-M27 May 2018 TZ2 · Paper 1 · Q11

11. What are the predicted electron domain geometries around the carbon and both nitrogen atoms in urea, $(NH_2)_2CO$, applying VSEPR theory?

	Carbon atom	Nitrogen atoms
A.	trigonal planar	trigonal pyramidal
B.	trigonal planar	tetrahedral
C.	tetrahedral	tetrahedral
D.	trigonal pyramidal	trigonal planar

3.1-M40 Nov 2012 · Paper 1 · Q12

12. In the molecule SF_4, which are the correct bond angles?



	α / °	β / °
A.	180	120
B.	187	103
C.	187	120
D.	180	90

Part B - Structured questions

3.1-S5 May 2024 TZ2 · Paper 2 · Q2 parts (a), (b)(i), (b)(ii), (b)(iii), (b)(iv), (c), (d) · also 2.1, 4.1, 4.2

2. In the stratosphere, ozone is decomposed by ultraviolet radiation.

$$O_3(g) \rightarrow O_2(g) + O(g)$$

(a) State the full electron configuration of an oxygen atom and the number of unpaired electrons in that atom. [2]

Electron configuration: . . . . . .

Unpaired electrons: . . . . . .

(b) (i) Draw a Lewis (electron dot) structure for the ozone molecule. [1]

(ii) Deduce the formal charge on each of the three oxygen atoms by adding them to your Lewis (electron dot) structure in (b)(i). [2]

(iii) Predict the shape and bond angle of the ozone molecule. [2]

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(iv) State the hybridization state of the central oxygen atom in ozone. [1]

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(c) Suggest a value, in pm, for the bond lengths in the ozone molecule and explain your answer. Use section 10 of the data booklet. [2]

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(d) Explain the dependence of the dissociation of diatomic oxygen, O_2, and ozone, O_3, on the wavelength of UV radiation. [2]

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★ **3.1-S14** Nov 2018 · Paper 2 · Q3 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (c), (d)(i), (d)(ii) · also 2.1, 6.2

3. Bromine can form the bromate(V) ion, BrO_3^-.

(a) (i) State the electron configuration of a bromine atom. [1]

. . . . . .

(ii) Sketch the orbital diagram of the **valence shell** of a bromine atom (ground state) on the energy axis provided. Use boxes to represent orbitals and arrows to represent electrons. [1]

(b) (i) Draw two Lewis (electron dot) structures for BrO_3^-. [2]

Structure I – follows octet rule:

Structure II – does not follow octet rule:

(c) Predict, using the VSEPR theory, the geometry of the BrO_3^- ion and the O–Br–O bond angles. [3]

Geometry:

. . . . . .

Reason:

. . . . . .

. . . . . .

O–Br–O angle:

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(d) (i) Bromate(V) ions act as oxidizing agents in acidic conditions to form bromide ions.

Deduce the half-equation for this reduction reaction. [2]

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(ii) Bromate(V) ions oxidize iron(II) ions, Fe^{2+}, to iron(III) ions, Fe^{3+}.

Deduce the equation for this redox reaction. [1]

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3.1-S24 May 2013 TZ2 · Paper 2 · Q5 parts (a)(iii), (a)(iv), (b)(i), (b)(ii), (c)(i), (c)(ii), (c)(iii), (c)(iv) · also 5.1, 9.2

5. Phosphoryl chloride, $POCl_3$, is a dehydrating agent.

(a) $POCl_3(g)$ decomposes according to the following equation.

$$2POCl_3(g) \rightarrow 2PCl_3(g) + O_2(g)$$

(iii) Define the term *standard enthalpy change of formation*, $\Delta H_f^{\ominus}$. *[1]*

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(iv) Calculate the standard enthalpy change for the reaction, $\Delta H^{\ominus}$, in kJ mol^{-1}, using the data below.

Substance	$\Delta H_f^{\ominus}$ / kJ mol^{-1}
$POCl_3(g)$	–542.2
$PCl_3(g)$	–288.1

[1]

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(b) (i) Deduce the Lewis (electron dot) structure of $POCl_3$ (with P as the central element) and PCl_3 and predict the shape of each molecule, using the valence shell electron pair repulsion theory (VSEPR). *[4]*

	$POCl_3$	PCl_3
Lewis (electron dot) structure		
Shape	. . . . .	. . . . .

(c) $POCl_3$ can be prepared by the reaction of phosphorus pentachloride, PCl_5, with tetraphosphorus decaoxide, P_4O_{10}.

(i) Deduce the Lewis (electron dot) structure of PCl_5. *[1]*

(ii) Predict the shape of this molecule, using the valence shell electron pair repulsion theory (VSEPR). *[1]*

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(iii) Identify all the different bond angles in PCl_5. *[1]*

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(iv) PCl_3Br_2 has the same molecular shape as PCl_5. Draw the three isomers of PCl_3Br_2 and deduce whether each isomer is polar or non-polar. *[3]*

	Isomer 1	Isomer 2	Isomer 3
Structure			
Molecular polarity	. . . . .	. . . . .	. . . . .

LESSON 4.1

Sigma & Pi Bonds and Delocalization

7 multiple-choice · 2 structured

Part A - Multiple choice

4.1-M2 Nov 2024 · Paper 1 · Q13

13. How many sigma (σ) and pi (π) bonds are present in the molecule below?

$(NC)_2CC(CN)_2$

	σ	π
A.	7	5
B.	9	5
C.	9	9
D.	13	5

4.1-M9 Nov 2022 · Paper 1 · Q9

9. Which molecule can be represented by resonance structures?

A. H_2S

B. HNO_3

C. H_2O_2

D. HClO

4.1-M13 May 2021 TZ1 · Paper 1 · Q12

12. Which contain delocalised electrons?

I. C_6H_5OH

II. CH_3COO^-

III. CO_3^{2-}

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

4.1-M15 Nov 2020 · Paper 1 · Q9

9. Which of these species contains the shortest carbon to oxygen bond length?

A. $CH_3CH_2O^-$

B. CH_3CH_2OH

C. CH_3COO^-

D. CH_3COOH

4.1-M25 May 2018 TZ2 · Paper 1 · Q13

13. Which overlap of atomic orbitals leads to the formation of only a sigma (σ) bond?

I. s – p
II. p – p
III. s – s

A. I and II only

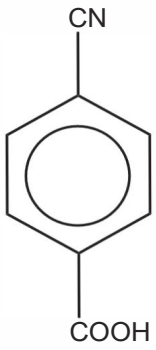
B. I and III only

C. II and III only

D. I, II and III

4.1-M26 Nov 2017 · Paper 1 · Q14

14. How many sigma (σ) and pi (π) bonds are present in this molecule?



	σ	π
A.	12	6
B.	14	5
C.	16	6
D.	17	5

4.1-M31 May 2017 TZ2 · Paper 1 · Q15

15. What can be deduced from the facts that ozone absorbs UV radiation in the region of 340 nm and molecular oxygen in the region of 242 nm?

A. The bond between atoms in molecular oxygen is a double bond.

B. The bonds in ozone are delocalized.

C. The bonds between atoms in ozone are stronger than those in molecular oxygen.

D. The bonds between atoms in molecular oxygen need more energy to break.

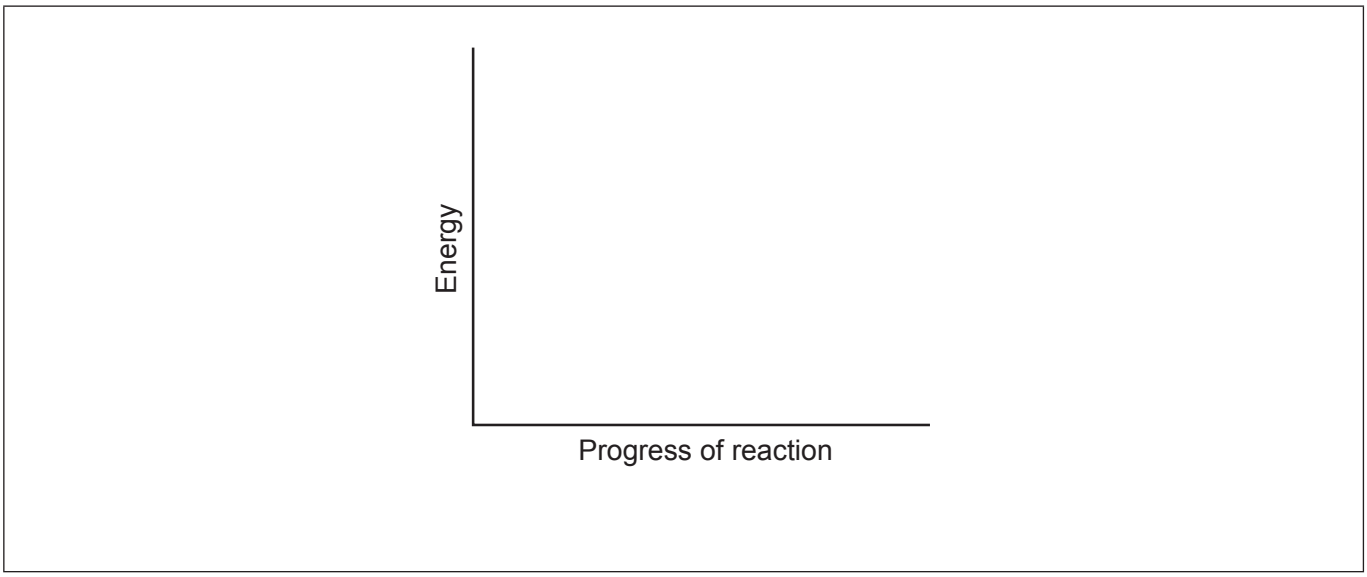
Part B - Structured questions

4.1-S11 May 2016 · Paper 2 · Q3 parts (b)(i), (c)(i), (c)(ii) · also 8.1

3. The reaction between hydrogen and nitrogen monoxide is thought to proceed by the mechanism shown below.

$2NO(g) \rightleftharpoons N_2O_2(g)$	fast equilibrium
$N_2O_2(g) + H_2(g) \rightarrow N_2O(g) + H_2O(g)$	slow reaction
$N_2O(g) + H_2(g) \rightarrow N_2(g) + H_2O(g)$	fast reaction

(b) The enthalpy change for the reaction between nitrogen monoxide and hydrogen is –664 kJ and its activation energy is 63 kJ.



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(ii) State what is meant by resonance. [1]

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★ **4.1-S4** Nov 2019 · Paper 2 · Q1 parts (a), (b), (c), (d), (e)(i), (e)(ii) · also 3.1

1. The equations show steps in the formation and decomposition of ozone in the stratosphere, some of which absorb ultraviolet light.

Step 1 $O_2 \rightarrow 2O\bullet$

Step 2 $O\bullet + O_2 \rightarrow O_3$

Step 3 $O_3 \rightarrow O\bullet + O_2$

Step 4 $O\bullet + O_3 \rightarrow 2O_2$

(a) Draw the Lewis structures of oxygen, O_2, and ozone, O_3. [2]

(b) Outline why both bonds in the ozone molecule are the same length and predict the bond length in the ozone molecule. Refer to section 10 of the data booklet. [2]

Reason:

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Length:

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(c) Predict the bond angle in the ozone molecule. [1]

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(d) Discuss how the different bond strengths between the oxygen atoms in O_2 and O_3 in the ozone layer affect radiation reaching the Earth's surface. [2]

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(e) (i) Identify the steps which absorb ultraviolet light. [1]

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(ii) Determine, showing your working, the wavelength, in m, of ultraviolet light absorbed by a single molecule in one of these steps. Use sections 1, 2 and 11 of the data booklet. [2]

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LESSON 4.2

Hybridization

4 multiple-choice · 1 structured

Part A - Multiple choice

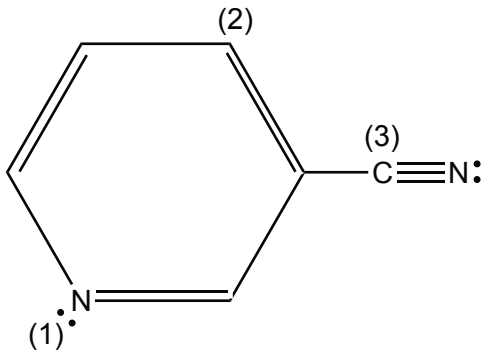
★ **4.2-M2** May 2025 TZ3 · Paper 1A · Q10

10. What is the hybridization of carbon and oxygen in CO_3^{2-}?

	C	O
A.	sp^2	sp^3
B.	sp^2	sp^2
C.	sp^3	sp^2
D.	sp^3	sp

★ **4.2-M3** Nov 2024 · Paper 1 · Q12

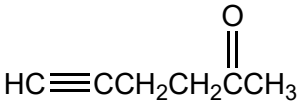
12. What is the molecular geometry and hybridization of the numbered atoms in the molecule shown below?



	N(1)	C(2)	C(3)
A.	sp^2 / bent	sp^2 / trigonal planar	sp / linear
B.	sp^2 / trigonal planar	sp^2 / bent	sp^2 / bent
C.	sp^3 / tetrahedral	sp^2 / trigonal planar	sp / bent
D.	sp / bent	sp^3 / tetrahedral	sp^3 / linear

4.2-M10 May 2019 TZ1 · Paper 1 · Q13

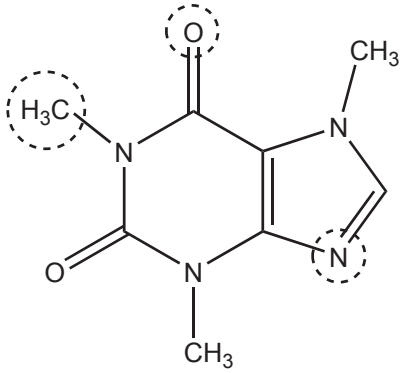
13. How many carbon atoms are sp^3, sp^2 and sp hybridized in the molecule?



	sp^3	sp^2	sp
A.	3	1	2
B.	2	1	3
C.	3	2	1
D.	3	2	2

4.2-M14 Nov 2017 · Paper 1 · Q13

13. What is the hybridization state and electron domain geometry around the circled C, N and O atoms?

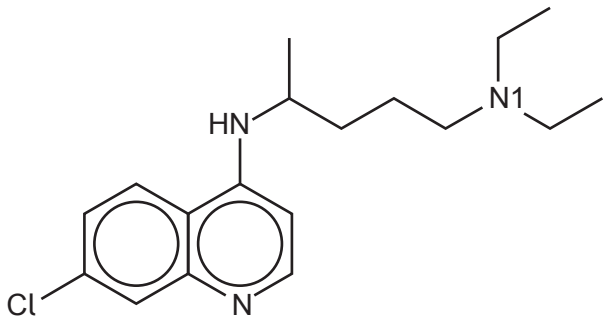


	C	O	N
A.	sp^3 and tetrahedral	sp^2 and trigonal planar	sp^2 and trigonal planar
B.	sp^2 and trigonal planar	sp and linear	sp^3 and tetrahedral
C.	sp^3 and tetrahedral	sp and linear	sp^2 and trigonal planar
D.	sp^3 and trigonal pyramidal	sp^2 and trigonal planar	sp^3 and trigonal pyramidal

Part B - Structured questions

4.2-S2 Nov 2022 · Paper 2 · Q2 parts (b), (d), (e)(ii) · also 2.1, 4.1

2. Chloroquine is a medication used to prevent and treat malaria.



(b) State the number of sp^2 hybridized carbons in chloroquine. [1]

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(d) Compare, giving a reason, the length of the carbon-nitrogen bond in the ring to the length of the carbon-N1 bond. [1]

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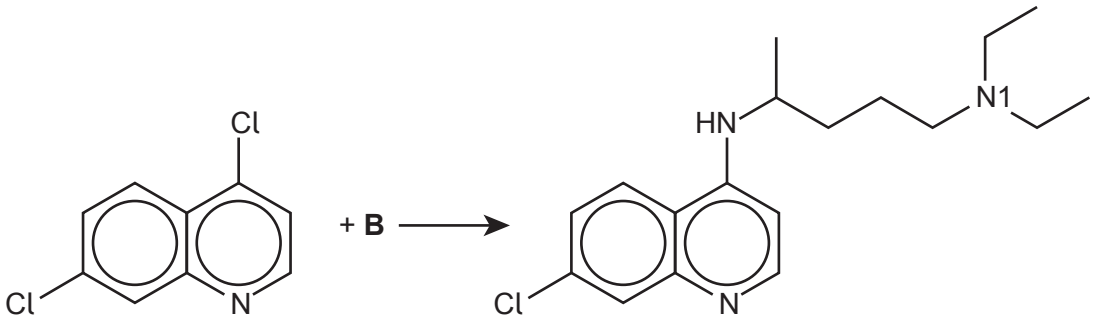
(e) Chloroquine can be synthesized by reacting 4,7-dichloroquinoline with another reactant, **B**.



4,7-dichloroquinoline **chloroquine**

(ii) This reaction can be done with a copper catalyst. State the ground-state electron configuration for copper. [1]

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LESSON 4.3

Covalent Network Structures

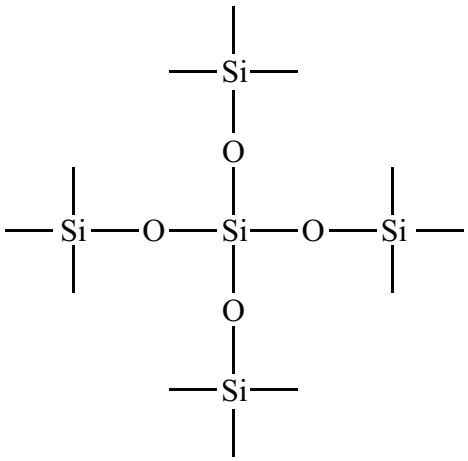
1 multiple-choice · 2 structured

Part A - Multiple choice

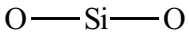
4.3-M10 May 2014 TZ2 · Paper 1 · Q13

13. Which diagram represents the bonding in SiO_2?

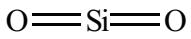
A.



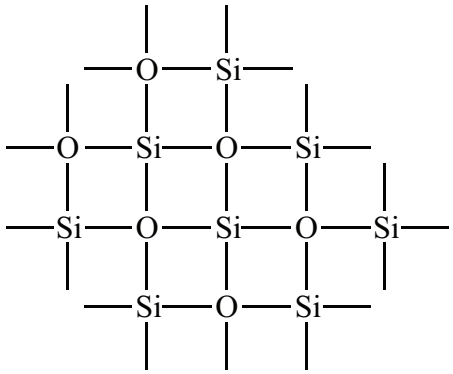
B.



C.



D.



Part B - Structured questions

★ **4.3-S6** May 2018 TZ2 · Paper 3 · Q1 parts (a)(i), (a)(ii), (b), (c)

1. The table summarizes some properties of graphite and graphene.

Property	Graphite	Graphene
Delocalization (Hybridization)	Yes (sp^2)	Yes (sp^2)
Electron mobility / $cm^2\ V^{-1}\ s^{-1}$	1800	15 000–200 000
Average bond length / nm	0.142	0.142
Distance between layers / nm	0.335	Not applicable (N/A)
Tensile strength / Pascals	$4.8–76 \times 10^6$	1.3×10^{11}
Density / $g\ cm^{-3}$	1.80–2.23	(N/A)
Melting point at 1×10^6 kPa / K	4300	4510
Specific surface area / $m^2\ g^{-1}$	90	2630

[Source: © Graphenea. Used with permission]

(a) (i) Graphene is two-dimensional, rather than three-dimensional, material.

Justify this by using the structure of graphene and information from the table. [2]

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(ii) Show that graphene is over 1600 times stronger than graphite. [1]

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(b) Diamond, graphene, and graphite are all network solids.

Suggest, giving a reason, the electron mobility of diamond compared to graphene. [2]

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(c) The melting point of diamond at 1×10^6 kPa is 4200 K (in the absence of oxygen).

Suggest, based on molecular structure, why graphene has a higher melting point under these conditions. [2]

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4.3-S10 May 2011 TZ1 · Paper 2 · Q6 parts (a), (b)(i), (b)(ii), (c), (d), (e) · also 4.1, 4.2

6. Carbon and silicon belong to the same group of the periodic table.

(a) Describe and compare **three** features of the structure and bonding in the three allotropes of carbon: diamond, graphite and C_{60} fullerene. *[6]*

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(b) Both silicon and carbon form oxides.

(i) Describe the structure and bonding in SiO_2. *[2]*

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(d) Describe the delocalization of pi (π) electrons and explain how this can account for the structure and stability of the carbonate ion, CO_3^{2-}. *[3]*

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(e) Explain the meaning of the term *hybridization*. State the type of hybridization shown by the carbon atoms in carbon dioxide, diamond, graphite and the carbonate ion. *[5]*

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LESSON 5.1

Molecular Polarity

2 multiple-choice · 0 structured

Part A - Multiple choice

5.1-M2 May 2024 TZ2 · Paper 1 · Q15

15. Which molecule is non-polar?

A. XeF_2

B. IF_5

C. SF_2

D. PF_3

★ **5.1-M3** Nov 2020 · Paper 1 · Q10

10. Which molecule is most polar?

A. CHF_3

B. CF_4

C. $CClF_3$

D. CCl_4

LESSON 5.2

Intermolecular Forces

5 multiple-choice · 2 structured

Part A - Multiple choice

★ **5.2-M3** Nov 2023 TZ1 · Paper 1 · Q11

11. Which substance exhibits only London (dispersion) forces between molecules?

A. PF_5

B. SF_4

C. SO_2

D. XeO_2

5.2-M4 May 2023 TZ2 · Paper 1 · Q10

10. What is the correct sequence if the compounds are arranged in order of increasing boiling point?

A. $CH_3OCH_3 < CH_3CH_2OH < CH_3CHO$

B. $CH_3OCH_3 < CH_3CHO < CH_3CH_2OH$

C. $CH_3CHO < CH_3CH_2OH < CH_3OCH_3$

D. $CH_3CHO < CH_3OCH_3 < CH_3CH_2OH$

5.2-M12 Nov 2018 · Paper 1 · Q11

11. Which compound has the highest boiling point?

A. CH_3CHO

B. CH_3CH_2F

C. CH_3OCH_3

D. $CH_3CH_2NH_2$

★ **5.2-M20** May 2015 TZ2 · Paper 1 · Q11

11. Which correctly lists butane ($M_r = 58$), propanone ($M_r = 58$), propan-1-ol ($M_r = 60$) and propan-2-ol ($M_r = 60$) in order of **increasing** boiling point?

A. $C_4H_{10} < CH_3COCH_3 < CH_3CH(OH)CH_3 < CH_3CH_2CH_2OH$

B. $CH_3CH_2CH_2OH < CH_3CH(OH)CH_3 < CH_3COCH_3 < C_4H_{10}$

C. $C_4H_{10} < CH_3CH(OH)CH_3 < CH_3CH_2CH_2OH < CH_3COCH_3$

D. $C_4H_{10} < CH_3COCH_3 < CH_3CH_2CH_2OH < CH_3CH(OH)CH_3$

5.2-M23 Nov 2013 · Paper 1 · Q11

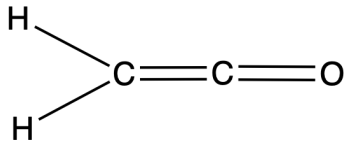
11. Which intermolecular forces exist between the following molecules?

	H_2Se	CO	H_2
A.	van der Waals' and dipole-dipole	van der Waals' and dipole-dipole	van der Waals' only
B.	van der Waals', dipole-dipole and hydrogen bonding	van der Waals' only	van der Waals' and hydrogen bonding
C.	van der Waals', dipole-dipole and hydrogen bonding	van der Waals' and dipole-dipole	van der Waals' only
D.	van der Waals' and dipole-dipole	van der Waals' and dipole-dipole	van der Waals' and hydrogen bonding

Part B - Structured questions

★ **5.2-S3** May 2024 TZ2 · Paper 2 · Q5 parts (b), (c)(i), (c)(ii) · also 8.1, 9.2

5. Ethenone, CH_2CO, is used in the synthesis of pharmaceutical compounds.



(b) Compare and contrast the intermolecular forces that result in ethenone being less volatile than carbon dioxide. [2]

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(c) Ethenone can be made by the thermal decomposition of propanone.

$$CH_3COCH_3\ (l) \rightarrow CH_2CO\ (g) + CH_4\ (g)$$

(i) Calculate the standard enthalpy change for this reaction.
Use $\Delta H^{\ominus}_f$ ethenone $= -87.2\,kJ\ mol^{-1}$ and section 12 of the data booklet. [2]

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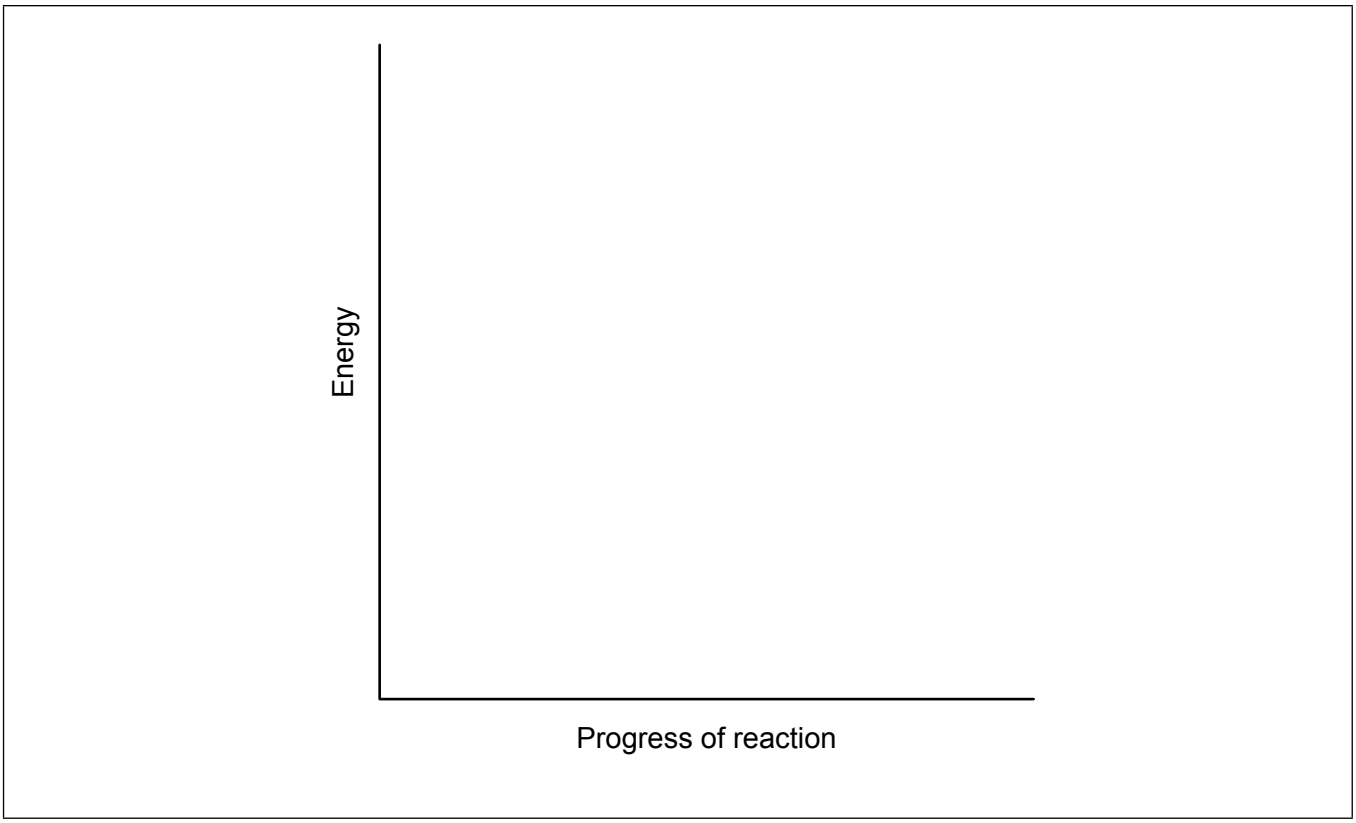
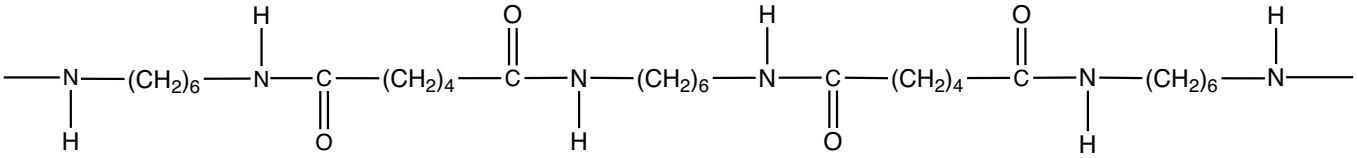
(ii) Sketch the potential energy diagram for the thermal decomposition of propanone from (c)(i). Use the axis given and indicate both the enthalpy of reaction and the activation energy. [2]



★ **5.2-S1** May 2025 TZ3 · Paper 2 · Q7 parts (b)(i), (b)(ii), (c) · also 4.3

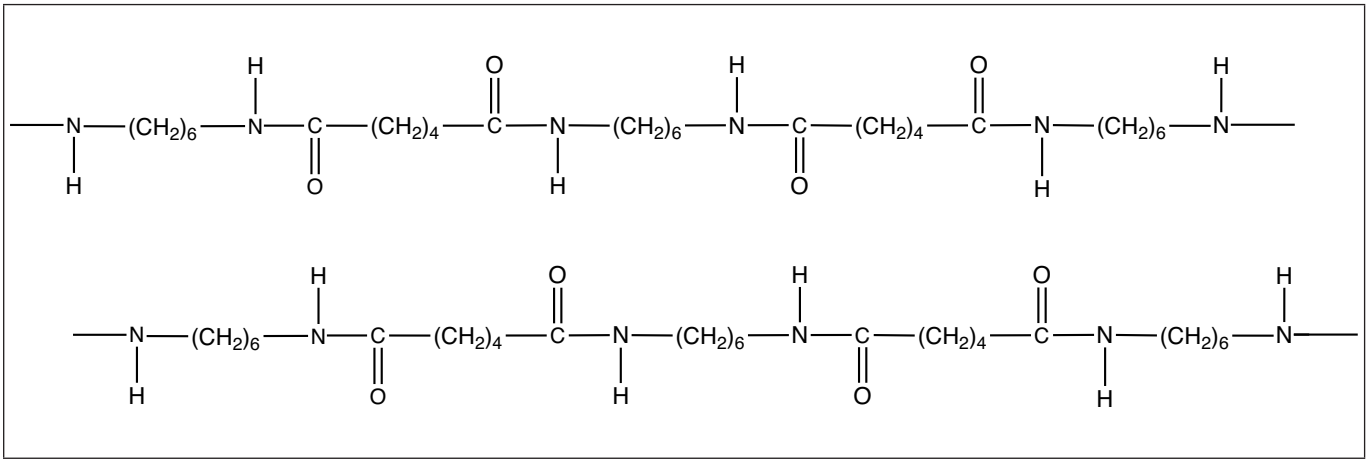
7. Nylon 6,6 is a polyamide used in many things, such as fishing lines.

A section of the polymer chain of nylon 6,6



(b) (i) The strength of nylon 6,6 is due to the hydrogen bonding between the polymer chains.

Annotate the diagram to show a hydrogen bond between the sections of the polymer chains. Use a dotted line to represent the hydrogen bond. [1]



(ii) Outline how a hydrogen bond is formed. [2]

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(c) Discuss the following data in terms of the bonding **and/or** intermolecular forces in each substance. [3]

Substance	**Melting point / K**
Silicon	1683
Poly(ethene)	393

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LESSON 5.3

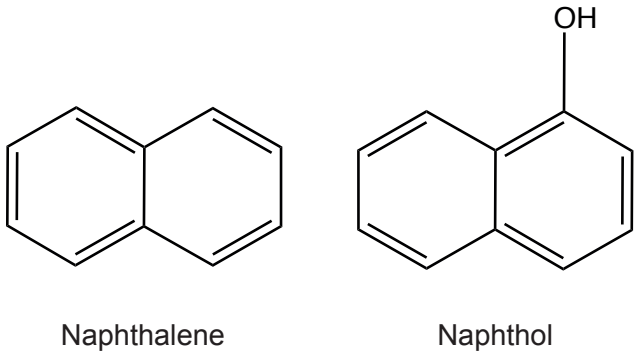
Separation Techniques

1 multiple-choice · 1 structured

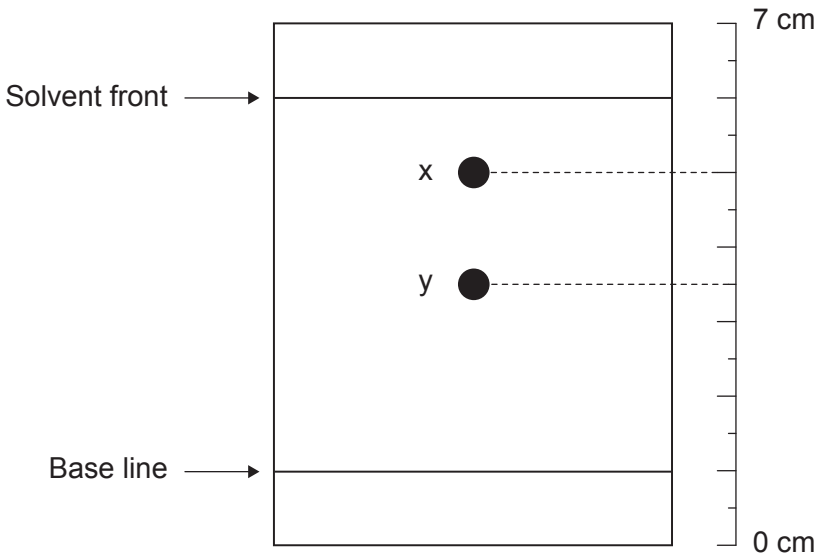
Part A - Multiple choice

★ **5.3-M1** May 2025 TZ1 · Paper 1A · Q12

12. The following thin-layer chromatogram was obtained from a mixture of naphthalene, $C_{10}H_8$, and naphthol, $C_{10}H_8O$. A polar silica was used as the stationary phase, and a non-polar hexane as the mobile phase.



Naphthalene Naphthol



What are the position and retardation factor (R_F) of naphthalene?

A. y and 0.50

B. y and 0.58

C. x and 0.80

D. x and 0.83

Part B - Structured questions

★ **5.3-S1** May 2025 TZ1 · Paper 1B · Q2 parts (a), (e)

2. A student was given a mixture to separate and collect the individual components.
The mixture contained sand, $SiO_2(s)$, sodium chloride, NaCl(s), and iron filings, Fe(s).
The student observed the original mixture and made the following hypothesis.

The iron would have the lowest percent by mass because it appeared to be present in the smallest quantity.

(a) Suggest a set of experimental steps required to obtain pure samples of each component of the mixture. [4]

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(e) In a different experiment, the students needed to separate a mixture of methanol and butan-1-ol. State the best method to separate this mixture and the physical property used for the separation. [2]

Separation method: . . . . . .

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Physical property: . . . . . .

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LESSON 6.1

Redox Reactions & Oxidation Numbers

9 multiple-choice · 2 structured

Part A - Multiple choice

★ **6.1-M6** Nov 2024 · Paper 1 · Q28

28. In which reaction does H_2 act as an oxidizing agent?

A. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

B. $2Na(s) + H_2(g) \rightarrow 2NaH(s)$

C. $Cl_2(g) + H_2(g) \rightarrow 2HCl(g)$

D. $H_2CCH_2(g) + H_2(g) \rightarrow H_3CCH_3(g)$

6.1-M10 May 2024 TZ2 · Paper 1 · Q32

32. Which of the following occurs during the Winkler test used to determine the concentration of dissolved oxygen in water?

Step I: $2Mn^{2+}(ag) + O_2(g) + 4OH^-(aq) \rightarrow 2MnO_2(s) + 2H_2O(l)$

Step II: $MnO_2(s) + 2I^-(aq) + 4H^+(aq) \rightarrow Mn^{2+}(aq) + I_2(aq) + 2H_2O(l)$

Step III: $2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$

A. Manganese(II) species is oxidized.

B. Iodine is reduced by oxygen to iodide ions.

C. Oxygen is oxidized.

D. Thiosulfate ions are reduced.

6.1-M12 May 2023 TZ2 · Paper 1 · Q28

28. Which species is the oxidizing agent?

$$14H^+(aq) + 2Mn^{2+}(aq) + 5BiO_3^-(aq) \rightarrow 2MnO_4^-(aq) + 5Bi^{3+}(aq) + 7H_2O(l)$$

A. $H^+(aq)$

B. $Mn^{2+}(aq)$

C. $BiO_3^-(aq)$

D. $MnO_4^-(aq)$

6.1-M13 May 2022 TZ1 · Paper 1 · Q28

28. In which of the following species would sulfur be reduced if converted to SCl_2?

A. $S_2O_3^{2-}$

B. H_2S

C. S

D. SO_2

6.1-M23 Nov 2018 · Paper 1 · Q8

8. Which is correct for the complex ion in $[Fe(H_2O)_5Cl]SO_4$?

	Oxidation state of iron	Overall charge of the complex ion
A.	+2	2+
B.	+2	0
C.	+3	1+
D.	+3	2+

★ **6.1-M24** Nov 2018 · Paper 1 · Q28

28. Which is correct for the reaction?

$$P_4(s) + 3H_2O(l) + 3OH^-(aq) \rightarrow PH_3(g) + 3H_2PO_2^-(aq)$$

	Oxidizing agent	Reducing agent
A.	H_2O	P_4
B.	P_4	OH^-
C.	OH^-	P_4
D.	P_4	P_4

★ **6.1-M25** May 2018 TZ1 · Paper 1 · Q28

28. Which equation shows oxygen undergoing reduction?

A. $2F_2 + O_2 \rightarrow 2F_2O$

B. $Na_2O + H_2O \rightarrow 2NaOH$

C. $H_2O_2 + 2HI \rightarrow 2H_2O + I_2$

D. $2CrO_4^{2-} + 2H^+ \rightleftharpoons Cr_2O_7^{2-} + H_2O$

6.1-M32 May 2017 TZ1 · Paper 1 · Q29

29. A reaction takes place when a rechargeable battery is used:

$$Pb(s) + PbO_2(s) + 4H^+(aq) + 2SO_4^{2-}(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$$

Which statements are correct?

I. H^+ is reduced
II. The oxidation state of Pb metal changes from 0 to +2
III. PbO_2 is the oxidising agent

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

6.1-M34 May 2017 TZ2 · Paper 1 · Q29

29. Which of the following is **not** a redox reaction?

A. $CH_4(g) + Cl_2(g) \rightarrow CH_3Cl(g) + HCl(g)$

B. $C(s) + O_2(g) \rightarrow CO_2(g)$

C. $2CO(g) \rightarrow CO_2(g) + C(s)$

D. $CH_3COOH(aq) + NaOH(aq) \rightarrow CH_3COONa(aq) + H_2O(l)$

Part B - Structured questions

★ **6.1-S3** May 2023 TZ1 · Paper 3 · Q2 parts (a)(i), (a)(ii), (b), (c) · also 6.3, 7.1, 7.2

2. Metals A, B, C, D, and E were placed in sulfate solutions.

(a) The results are given where ✓ = reaction occurred and × = no reaction.

Metal	$ASO_4(aq)$	$BSO_4(aq)$	$CSO_4(aq)$	$DSO_4(aq)$	$ESO_4(aq)$
A	—	✓	×	✓	✓
B	×	—	×	✓	✓
C	✓	✓	—	✓	✓
D	×	×	×	—	✓
E	×	×	×	×	—

(i) Identify the oxidation state of metal A in ASO_4. [1]

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(ii) Deduce the activity series of the metals. [1]

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(b) Sketch the voltaic cell that produces the greatest electromotive force, EMF, using the information in part (a). Label the direction of electron flow, salt bridge, anode, cathode, the metals (A–E) and solutions used in each half-cell. [4]

(c) Deduce **one** metal that could be electroplated, with high efficiency, on to a metal key using an aqueous salt of the metal. Use section 24 of the data booklet. [1]

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★ **6.1-S14** Nov 2012 · Paper 2 · Q4 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i), (b)(ii), (b)(iii) · also 6.2, 7.2

4. Arsenic and nitrogen play a significant role in environmental chemistry. Arsenous acid, H_3AsO_3, can be found in oxygen-poor (anaerobic) water, and nitrogen-containing fertilizers can contaminate water.

(a) (i) Define *oxidation* and *reduction* in terms of electron loss or gain. *[1]*

Oxidation:

. . . . . .

Reduction:

. . . . . .

(ii) Deduce the oxidation numbers of arsenic and nitrogen in each of the following species. *[4]*

As_2O_3: . . . . . .

NO_3^-: . . . . . .

H_3AsO_3: . . . . . .

N_2O_3: . . . . . .

(iii) Distinguish between the terms *oxidizing agent* and *reducing agent*. *[1]*

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(iv) In the removal of arsenic from contaminated groundwater, H_3AsO_3 is often first oxidized to arsenic acid, H_3AsO_4.

The following **unbalanced** redox reaction shows another method of forming H_3AsO_4.

$$As_2O_3(s) + NO_3^-(aq) \rightarrow H_3AsO_4(aq) + N_2O_3(aq)$$

Deduce the balanced redox equation in **acid**, and then identify both the oxidizing and reducing agents. *[3]*

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(b) The electrolysis of aqueous copper(II) sulfate is an example of an electrolysis process where the nature of the electrodes can determine which products form. Platinum electrodes were used in **process 1** and copper electrodes in **process 2**.

(i) Draw an annotated diagram of the electrolytic cell in **process 1** and identify the direction of electron flow. *[2]*

(ii) For **process 1** (platinum electrodes), state the half-equations occurring at the positive electrode (anode) and negative electrode (cathode). **Include state symbols for all species**. Describe what is observed at each electrode and comment on any change in the colour and the acidity of the solution. *[7]*

Half-equation at positive electrode (anode):

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Half-equation at negative electrode (cathode):

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Observation at positive electrode (anode):

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Observation at negative electrode (cathode):

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Change in colour (if any) of the solution:

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Change in acidity (if any) of the solution:

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(iii) For **process 2** (copper electrodes), state the half-equations occurring at the positive electrode (anode) and negative electrode (cathode). **Include state symbols for all species**. Describe what is observed at each electrode and comment on any change in the colour and the acidity of the solution. *[7]*

Half-equation at positive electrode (anode):

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Half-equation at negative electrode (cathode):

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Observation at positive electrode (anode):

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Observation at negative electrode (cathode):

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Change in colour (if any) of the solution:

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Change in acidity (if any) of the solution:

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LESSON 6.2

Balancing Redox Reactions

2 multiple-choice · 1 structured

Part A - Multiple choice

★ **6.2-M1** Nov 2022 · Paper 1 · Q29

29. What is the coefficient for H^+ when the equation below is balanced?

$$__Pb(s) + __NO_3^-(aq) + __H^+(aq) \rightarrow __Pb^{2+}(aq) + __NO(g) + __H_2O(l)$$

A. 2

B. 4

C. 6

D. 8

6.2-M3 May 2015 TZ2 · Paper 1 · Q31

31. What is the coefficient for I^- when the following equation is balanced using the smallest possible whole numbers?

$$IO_3^-(aq) + ___ I^-(aq) + ___ H^+(aq) \rightarrow ___ I_2(aq) + ___ H_2O(l)$$

A. 1

B. 2

C. 3

D. 5

Part B - Structured questions

★ **6.2-S3** May 2017 TZ2 · Paper 2 · Q2 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i) · also 1.2, 7.1

2. (a) An acidic sample of a waste solution containing $Sn^{2+}(aq)$ reacted completely with $K_2Cr_2O_7$ solution to form $Sn^{4+}(aq)$.

(i) State the oxidation half-equation. [1]

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(ii) Deduce the overall redox equation for the reaction between acidic $Sn^{2+}(aq)$ and $Cr_2O_7^{2-}(aq)$, using section 24 of the data booklet. [1]

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(iii) 13.239 g of $K_2Cr_2O_7(s)$ were dissolved in distilled water to form a 0.100 dm^3 solution. Calculate its molar concentration. [1]

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(iv) 10.0 cm^3 of the waste sample required 13.24 cm^3 of the $K_2Cr_2O_7$ solution. Calculate the molar concentration of $Sn^{2+}(aq)$ in the waste sample. [2]

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(b) (i) Corrosion of iron is similar to the processes that occur in a voltaic cell. The initial steps involve the following half-equations:

$$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$$

$$\frac{1}{2}O_2(g) + H_2O(l) + 2e^- \rightleftharpoons 2OH^-(aq)$$

Calculate $E^{\ominus}$, in V, for the spontaneous reaction using section 24 of the data booklet. [1]

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LESSON 6.3

Reactivity

2 multiple-choice · 2 structured

Part A - Multiple choice

★ **6.3-M3** Nov 2023 TZ1 · Paper 1 · Q29

29. The acid H_2S reacts with an active metal, M. Which combination shows the correct role of H_2S, and product formed from the reaction?

	Role of H_2S	Product from H_2S reaction
A.	Oxidizing agent	$H_2(g)$
B.	Oxidizing agent	$S(s)$
C.	Reducing agent	$H_2(g)$
D.	Reducing agent	$S(s)$

6.3-M4 May 2021 TZ1 · Paper 1 · Q28

28. A student performed displacement reactions using metals W and X and solutions of salts of metals W, X, Y and Z. The results are summarized in the table.

		Salt solution			
		W^{2+}	X^{2+}	Y^{2+}	Z^{2+}
Metal	W		No reaction	No reaction	No reaction
	X	Reaction		Reaction	No reaction

Which of the four metals is most reactive?

A. W

B. X

C. Y

D. Z

Part B - Structured questions

6.3-S3 Nov 2015 · Paper 2 · Q7 parts (d)(i), (d)(ii), (e) · also 7.2

(d) Consider the following spontaneous reactions.

$$Fe(s) + Ni^{2+}(aq) \rightarrow Fe^{2+}(aq) + Ni(s)$$

$$Zn(s) + Fe^{2+}(aq) \rightarrow Zn^{2+}(aq) + Fe(s)$$

$$Ni(s) + Pb^{2+}(aq) \rightarrow Ni^{2+}(aq) + Pb(s)$$

(i) Deduce the order of **increasing** reactivity of the metals based on the reactions above. [2]

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(ii) Identify the strongest oxidizing agent in the reactions above. [1]

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(e) Deduce the half-equations for the formation of the major product at the positive electrode (anode) when the following aqueous solutions are electrolysed. [2]

Dilute sodium chloride:

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Concentrated sodium chloride:

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6.3-S1 May 2022 TZ1 · Paper 2 · Q2 parts (a), (b)(i), (b)(iii) · also 7.1, 7.2

2. Magnesium is a reactive metal often found in alloys.

(a) Suggest an experiment that shows that magnesium is more reactive than zinc, giving the observation that would confirm this. [2]

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(b) Magnesium is sometimes used as a sacrificial anode to protect steel from corrosion.

(i) Calculate the standard potential, in V, of a cell formed by magnesium and steel half-cells. Use section 24 of the data booklet and assume steel has the standard electrode potential of iron. [1]

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(iii) This cell causes the electrolytic reduction of water on the steel. State the half-equation for this reduction. [1]

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LESSON 7.1

Primary & Secondary Cells

9 multiple-choice · 5 structured

Part A - Multiple choice

7.1-M8 May 2023 TZ1 · Paper 1 · Q31

31. Which statement is correct about the ions in a cell assembled from these half-cells?

Reaction	$E^{\ominus}$
$Ni^{2+}(aq) + 2e^{-} \rightleftharpoons Ni(s)$	$-0.26\,V$
$Zn^{2+}(aq) + 2e^{-} \rightleftharpoons Zn(s)$	$-0.76\,V$

A. Negative ions flow into the zinc half-cell from the salt bridge.

B. Negative ions flow into the nickel half-cell from the salt bridge.

C. Zn^{2+} ions are reduced to Zn.

D. The concentration of Ni^{2+} ions increases.

7.1-M12 Nov 2022 · Paper 1 · Q30

30. The standard electrode potentials for three half-cells involving chromium are shown.

$$Cr^{3+}(aq) + e^{-} \rightleftharpoons Cr^{2+}(aq) \qquad E^{\ominus} = -0.407\,V$$

$$Cr^{3+}(aq) + 3e^{-} \rightleftharpoons Cr(s) \qquad E^{\ominus} = -0.744\,V$$

$$Cr^{2+}(aq) + 2e^{-} \rightleftharpoons Cr(s) \qquad E^{\ominus} = -0.914\,V$$

Which statement is correct?

A. $Cr^{3+}(aq)$ can oxidize $Cr^{2+}(aq)$ but not $Cr(s)$.

B. $Cr^{3+}(aq)$ can oxidize $Cr(s)$ but not $Cr^{2+}(aq)$.

C. $Cr^{3+}(aq)$ can oxidize both $Cr^{2+}(aq)$ and $Cr(s)$.

D. $Cr^{3+}(aq)$ can oxidize $Cr(s)$ and reduce $Cr^{2+}(aq)$.

7.1-M29 May 2016 · Paper 1 · Q31

31. Which statement is correct for a voltaic but **not** for an electrolytic cell?

A. An electrolyte is required.

B. The anode is where oxidation occurs.

C. Ions move in the electrolyte.

D. Electrons flow from the negative electrode to the positive electrode.

★ **7.1-M14** May 2022 TZ2 · Paper 1 · Q30

30. Which $E^{\ominus}$ value, in V, for the reaction $Mn(s) + Zn^{2+}(aq) \rightarrow Mn^{2+}(aq) + Zn(s)$ can be deduced from the following equations?

$$Mn(s) + 2Ag^{+}(aq) \rightarrow Mn^{2+}(aq) + 2Ag(s) \qquad E^{\ominus} = 1.98\,V$$

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s) \qquad E^{\ominus} = 1.10\,V$$

$$Cu(s) + 2Ag^{+}(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s) \qquad E^{\ominus} = 0.46\,V$$

A. 0.42

B. 1.34

C. 2.62

D. 3.54

7.1-M18 May 2021 TZ2 · Paper 1 · Q30

30. What would be the electrode potential, $E^{\ominus}$, of the $Mn^{2+}(aq)|Mn(s)$ half-cell if $Fe^{3+}(aq)|Fe^{2+}(aq)$ is used as the reference standard?

$$Mn^{2+}(aq) + 2e^{-} \rightleftharpoons Mn(s) \qquad E^{\ominus} = -1.18\,V$$

$$Fe^{3+}(aq) + e^{-} \rightleftharpoons Fe^{2+}(aq) \qquad E^{\ominus} = +0.77\,V$$

A. −1.95 V

B. −0.41 V

C. +0.41 V

D. +1.95 V

★ **7.1-M30** Nov 2015 · Paper 1 · Q32

32. What is $E^{\ominus}$, in V, for the following reaction?

$$VO^{2+}(aq) + 2H^{+}(aq) + V^{2+}(aq) \rightarrow 2V^{3+}(aq) + H_2O(l)$$

	Standard electrode potential, $E^{\ominus}$ / V
$V^{2+}(aq) + 2e^{-} \rightleftharpoons V(s)$	−1.18
$V^{3+}(aq) + e^{-} \rightleftharpoons V^{2+}(aq)$	−0.26
$VO^{2+}(aq) + 2H^{+}(aq) + e^{-} \rightleftharpoons V^{3+}(aq) + H_2O(l)$	+0.34
$VO_2^{+}(aq) + 2H^{+}(aq) + e^{-} \rightleftharpoons VO^{2+}(aq) + H_2O(l)$	+1.00

A. −0.60

B. +0.08

C. +0.60

D. +1.26

7.1-M33 May 2015 TZ2 · Paper 1 · Q32

32. The standard electrode potentials for three reactions involving copper and copper ions are:

$$Cu^{2+}(aq) + e^- \rightleftharpoons Cu^+(aq) \qquad E^\ominus = +0.15\,V$$
$$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s) \qquad E^\ominus = +0.34\,V$$
$$Cu^+(aq) + e^- \rightleftharpoons Cu(s) \qquad E^\ominus = +0.52\,V$$

Which statement is correct?

A. Cu^{2+} ions are a better oxidizing agent than Cu^+ ions.

B. Copper metal is a better reducing agent than Cu^+ ions.

C. Cu^+ ions will spontaneously form copper metal and Cu^{2+} ions in solution.

D. Copper metal can be spontaneously oxidized by Cu^{2+} ions to form Cu^+ ions.

7.1-M41 May 2013 TZ1 · Paper 1 · Q33

33. The standard electrode potentials of some half-reactions are given below.

$$Sn^{4+}(aq) + 2e^- \rightleftharpoons Sn^{2+}(aq) \qquad E^\ominus = +0.15\ V$$
$$\tfrac{1}{2}I_2(s) + e^- \rightleftharpoons I^-(aq) \qquad E^\ominus = +0.54\ V$$
$$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq) \qquad E^\ominus = +0.77\ V$$

Which of the following reactions will occur spontaneously?

A. Iodine reduces Fe^{3+} to Fe^{2+}

B. Iodine reduces Sn^{4+} to Sn^{2+}

C. Iodine oxidizes Fe^{2+} to Fe^{3+}

D. Iodine oxidizes Sn^{2+} to Sn^{4+}

7.1-M56 May 2010 TZ2 · Paper 1 · Q31

31. Which changes could take place at the positive electrode (cathode) in a voltaic cell?

I. $Zn^{2+}(aq)$ to $Zn(s)$

II. $Cl_2(g)$ to $Cl^-(aq)$

III. $Mg(s)$ to $Mg^{2+}(aq)$

A. I and II only

B. I and III only

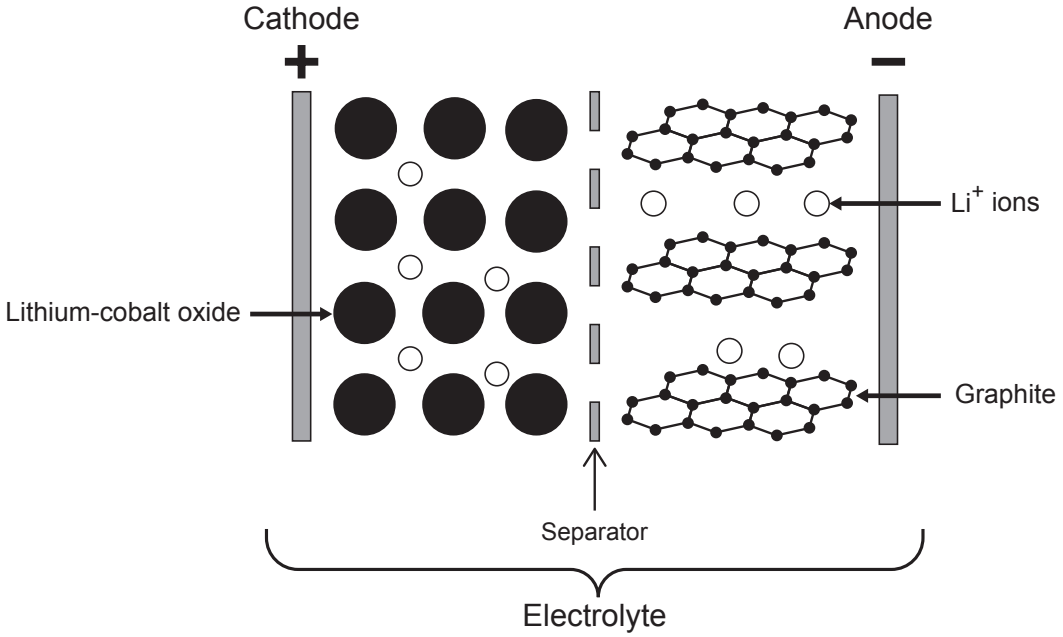
C. II and III only

D. I, II and III

Part B - Structured questions

7.1-S1 May 2025 TZ3 · Paper 2 · Q6 parts (a)(i), (a)(ii), (b)

6. A lithium-ion battery consists of lithium ions on lithium-graphite (LiC_6) and lithium-cobalt oxide ($LiCoO_2$) in an electrolyte.



Lithium-ion battery

(a) The lithium ions are produced at the anode from the lithium atoms embedded in graphite.

(i) State a half-equation for the reaction occurring at the anode during discharge. [1]

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(ii) Comment on the sign and value of the standard reduction potential of lithium that make it suitable to use in the battery. Use section 19 of the data booklet. [2]

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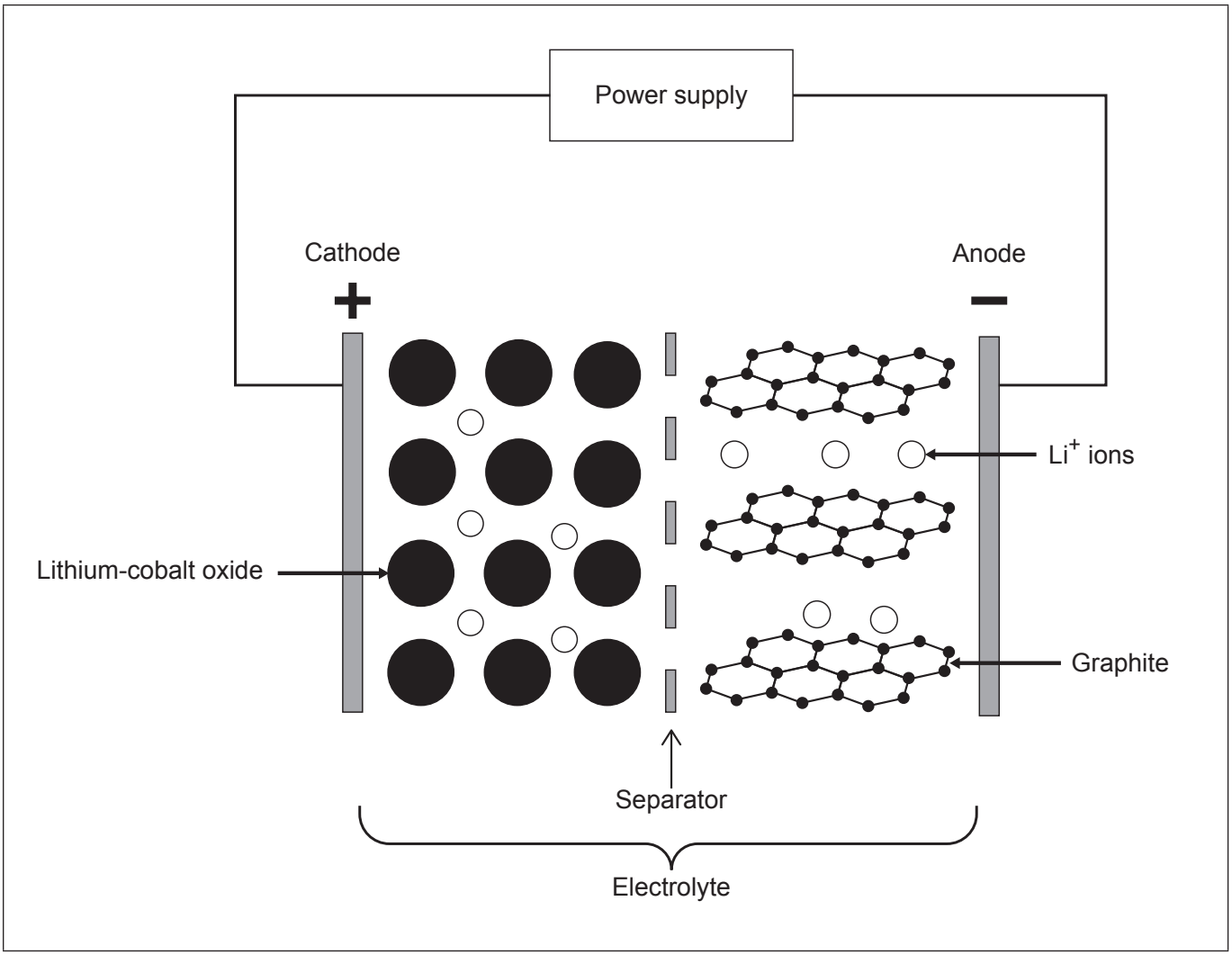
(b) The lithium-ion battery is rechargeable. Annotate the diagram below to show the direction of the flow of electrons in the wire while the battery is recharging. [1]



★ **7.1-S6** May 2024 TZ2 · Paper 3 · Q13 parts (a), (b)(i), (c)(i), (c)(ii), (c)(iii) · also 7.3

13. Batteries and fuels provide portable sources of energy.

(a) Suggest, with a reason, if specific energy or energy density is a better measure of a fuel's usefulness as an everyday portable energy source. [1]

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(b) (i) Ethylbenzene, $C_6H_5CH_2CH_3$, is an aromatic compound that is used to increase the octane rating in petrol (gasoline). It has a specific energy of $4.135 \times 10^7\,J\,kg^{-1}$.

Calculate the enthalpy of combustion of ethylbenzene, in $kJ\,mol^{-1}$, using section 6 of the data booklet. [2]

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(c) (i) Lithium ion and lead acid are both types of rechargeable batteries.

Outline the mass and voltage advantages of a lithium-ion battery, using sections 6 and 24 of the data booklet. [2]

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(ii) Graphite and a lithium cobalt oxide complex, $LiCoO_2$ (s), are the electrodes used in a lithium-ion cell.

Deduce the half-equations for the reactions occurring at each electrode during the charging of a lithium-ion cell. [2]

Graphite:

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$LiCoO_2$ (s):

. . . . . .

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(iii) State **one** disadvantage of the lithium-ion battery. [1]

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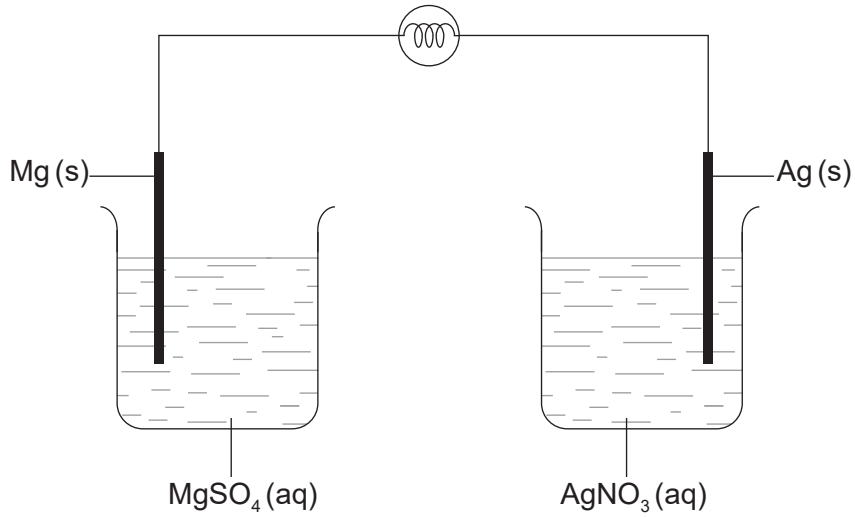
7.1-S18 May 2018 TZ1 · Paper 2 · Q6 parts (a), (b), (c), (d), (e) · also 1.2

6. The diagram shows an incomplete voltaic cell with a light bulb in the circuit.



(a) Identify the missing component of the cell and its function. [2]

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(b) Deduce the half-equations for the reaction at each electrode when current flows. [2]

Positive electrode (cathode):

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Negative electrode (anode):

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(c) Annotate the diagram with the location and direction of electron movement when current flows. [1]

(d) Calculate the cell potential, in V, using section 24 of the data booklet. [1]

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(e) Determine the loss in mass of one electrode if the mass of the other electrode increases by 0.10 g. [2]

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★ **7.1-S21** May 2017 TZ1 · Paper 2 · Q3 parts (a), (b)(i), (b)(ii), (c)(i) · also 6.1, 6.2

3. Vanadium has a number of different oxidation states.

(a) Determine the oxidation state of vanadium in each of the following species. [2]

V_2O_5:

. . . . . .

VO^{2+}:

. . . . . .

(ii) Identify, from the table, a non-vanadium species that could convert $VO_2^+(aq)$ to $V^{2+}(aq)$. [1]

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(c) (i) Formulate an equation for the reaction between $VO^{2+}(aq)$ and $V^{2+}(aq)$ in acidic solution to form $V^{3+}(aq)$. [1]

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★ **7.1-S29** Nov 2013 · Paper 2 · Q6 parts (a), (b)(i), (c), (d)(i), (d)(ii), (d)(iii), (e)(iii), (f)(i), (f)(ii) · also 3.1, 4.2, 6.1

6. In acidic solution, ions containing titanium can react according to the half-equation below.

$$TiO^{2+}(aq) + 2H^+(aq) + e^- \rightleftharpoons Ti^{3+}(aq) + H_2O(l) \qquad E^\ominus = -0.06\,V$$

(a) Define the term *standard electrode potential*, $E^\ominus$. *[1]*

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(b) (i) State the initial and final oxidation numbers of titanium and hence deduce whether it is oxidized or reduced in this change. *[2]*

Initial oxidation number	Final oxidation number	Oxidized / reduced

(c) In the two experiments below, predict whether a reaction would occur and deduce an equation for any reaction that takes place. Refer to Table 14 of the Data Booklet if necessary. *[3]*

KI(aq) is added to a solution containing Ti^{3+}(aq) ions:

. . . . . .

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Zn(s) is added to a solution containing TiO^{2+}(aq) and H^{+}(aq) ions:

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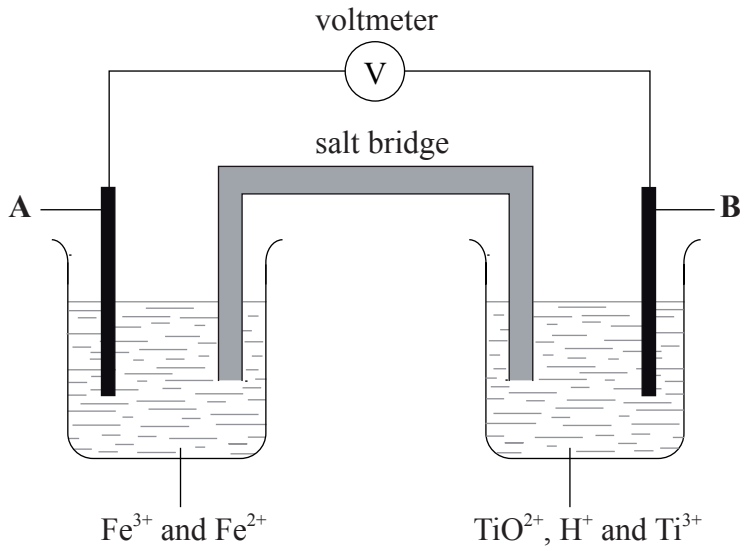
(d) In the diagram below, **A** and **B** are inert electrodes and, in the aqueous solutions, all ions have a concentration of 1 mol dm^{-3}.



(i) Using Table 14 of the Data Booklet, state the balanced half-equation for the reaction that occurs at electrode **A** and whether it involves oxidation or reduction. *[2]*

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(ii) Calculate the cell potential in V. *[1]*

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(iii) On the diagram above label with an arrow

- the direction of electron flow in the wire
- the direction in which the positive ions flow in the salt bridge. *[1]*

(e) Sodium, silicon and sulfur are elements in period 3 of the periodic table that all form oxides.

(iii) As well as the oxide above, sodium forms a peroxide that contains the peroxide ion, O_2^{2-}. Draw the Lewis (electron dot) structure of the peroxide ion. *[2]*

(f) Although carbon and silicon both belong to group 4 of the periodic table, carbon dioxide and silicon dioxide are different in many ways.

(i) Describe the differences in the hybridization of these group 4 elements and the precise nature of the bonds that they form with the oxygen atoms. *[3]*

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(ii) Xenon, although a noble gas, forms an oxide, XeO_2, that has a structure related to that of SiO_2. Compare the geometry around the silicon atoms in SiO_2 with the geometry around the xenon atoms in XeO_2, using the valence shell electron pair repulsion (VSEPR) theory. *[3]*

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LESSON 7.2

Electrolytic Cells

4 multiple-choice · 3 structured

Part A - Multiple choice

★ **7.2-M6** May 2023 TZ2 · Paper 1 · Q31

31. Which will eventually yield the greatest mass of deposited copper in the electrolysis of a fixed volume of 1 mol dm^{-3} $CuSO_4(aq)$?

A. copper anode and inert cathode

B. inert anode and copper cathode

C. inert anode and inert cathode

D. zinc anode and zinc cathode

★ **7.2-M8** Nov 2021 · Paper 1 · Q31

31. Which aqueous solutions produce oxygen gas during electrolysis?

I. Dilute $CuCl_2(aq)$ with inert electrodes

II. Dilute $FeSO_4(aq)$ with inert electrodes

III. Dilute $CuCl_2(aq)$ with copper electrodes

The standard electrode potentials are provided in the table:

Half-equation	$E^{\ominus}$ / V
$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$	−0.45
$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$	+0.34
$\frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightleftharpoons H_2O(l)$	+1.23
$\frac{1}{2}Cl_2(g) + e^- \rightleftharpoons Cl^-(aq)$	+1.36

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

7.2-M9 May 2021 TZ1 · Paper 1 · Q31

31. What are the products when concentrated aqueous copper(II) chloride is electrolysed using platinum electrodes?

	Anode (positive electrode)	Cathode (negative electrode)
A.	$O_2(g)$	$Cu(s)$
B.	$Cl_2(g)$	$H_2(g)$
C.	$Cl_2(g)$	$Cu(s)$
D.	$O_2(g)$	$H_2(g)$

7.2-M26 Nov 2013 · Paper 1 · Q34

34. What happens during the electrolysis of concentrated aqueous potassium chloride?

I. Reduction takes place at the negative electrode (cathode).

II. Hydrogen gas is evolved at the negative electrode (cathode).

III. The pH of the electrolyte increases.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

Part B - Structured questions

★ **7.2-S19** May 2014 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(ii), (a)(v), (b)(i), (b)(ii), (b)(iii), (b)(iv), (c)(i), (c)(ii), (d)(ii), (d)(iii) · also 6.2, 6.3, 7.1, 7.3

6. Oxidation and reduction can be defined in terms of electron transfer or oxidation numbers.

(a) Alcohols with the molecular formula C_4H_9OH occur as four structural isomers. Three of the isomers can be oxidized with acidified potassium dichromate solution to form compounds with the molecular formula C_4H_8O.

(i) Deduce the half-equation for the oxidation of the alcohol C_4H_9OH. *[1]*

. . . . . .

(ii) Deduce the overall equation for the redox reaction. *[1]*

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(v) All isomers of the alcohol C_4H_9OH undergo complete combustion. State an equation for the complete combustion of C_4H_9OH. *[2]*

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. . . . . .

(b) A reactivity series can be experimentally determined by adding the metals W, X, Y and Z to solutions of these metal ions. The following reactions were observed:

$$W^{2+}(aq) + X(s) \rightarrow W(s) + X^{2+}(aq)$$
$$Y(s) + W^{2+}(aq) \rightarrow Y^{2+}(aq) + W(s)$$
$$Z^{2+}(aq) + W(s) \rightarrow Z(s) + W^{2+}(aq)$$
$$Y(s) + X^{2+}(aq) \rightarrow Y^{2+}(aq) + X(s)$$

(i) Deduce the order of reactivity of these four metals, from the least to the most reactive. *[1]*

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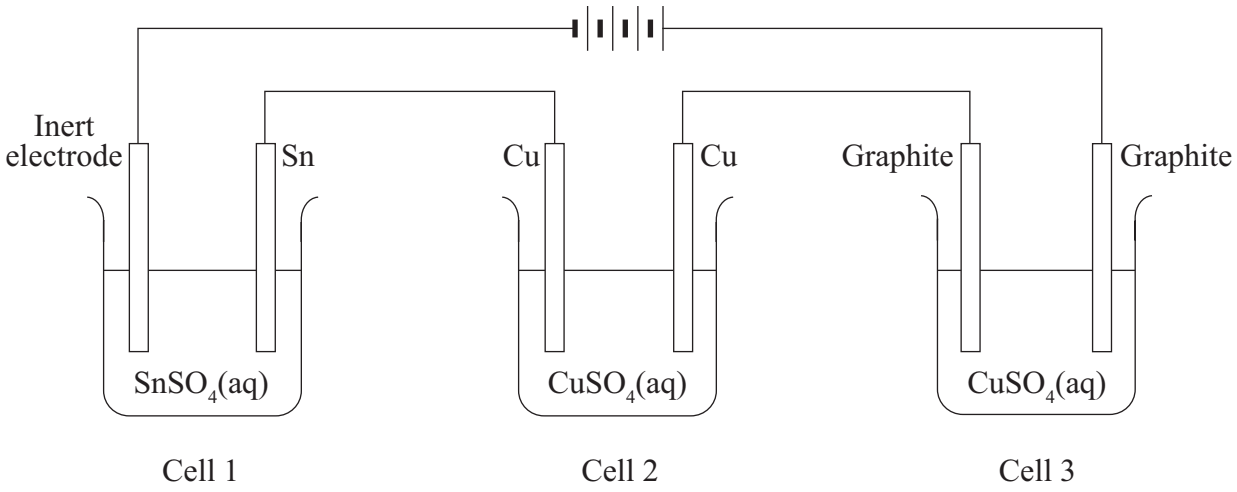
(c) A student carries out the electrolysis of aqueous potassium iodide, KI, using inert electrodes.

(d) Three electrolytic cells were set up in series (one cell after the other), as shown below. All of the solutions had a concentration of $1.00\,mol\,dm^{-3}$.



(ii) Compare the colour and the pH of the solutions in cells 2 and 3 after the current has been flowing for one hour. *[2]*

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(iii) Explain your answer given for part (d) (ii). *[2]*

Colour:

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pH:

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7.2-S7 Nov 2019 · Paper 2 · Q6 parts (a)(i), (a)(ii), (b), (c)(i), (c)(ii), (c)(iii), (c)(iv), (e)(i), (e)(ii) · also 1.2, 2.1, 7.1, 9.2

6. Copper forms two chlorides, copper(I) chloride and copper(II) chloride.

(a) (i) State the electron configuration of the Cu^+ ion. [1]

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(ii) Copper(II) chloride is used as a catalyst in the production of chlorine from hydrogen chloride.

$$4HCl(g) + O_2(g) \rightarrow 2Cl_2(g) + 2H_2O(g)$$

Calculate the standard enthalpy change, $\Delta H^{\ominus}$, in kJ, for this reaction, using section 12 of the data booklet. [2]

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(b) Solid copper(II) chloride absorbs moisture from the atmosphere to form a hydrate of formula $CuCl_2 \bullet xH_2O$.

A student heated a sample of hydrated copper(II) chloride, in order to determine the value of x. The following results were obtained:

Mass of crucible = 16.221 g
Initial mass of crucible and hydrated copper(II) chloride = 18.360 g
Final mass of crucible and anhydrous copper(II) chloride = 17.917 g

Determine the value of x. [3]

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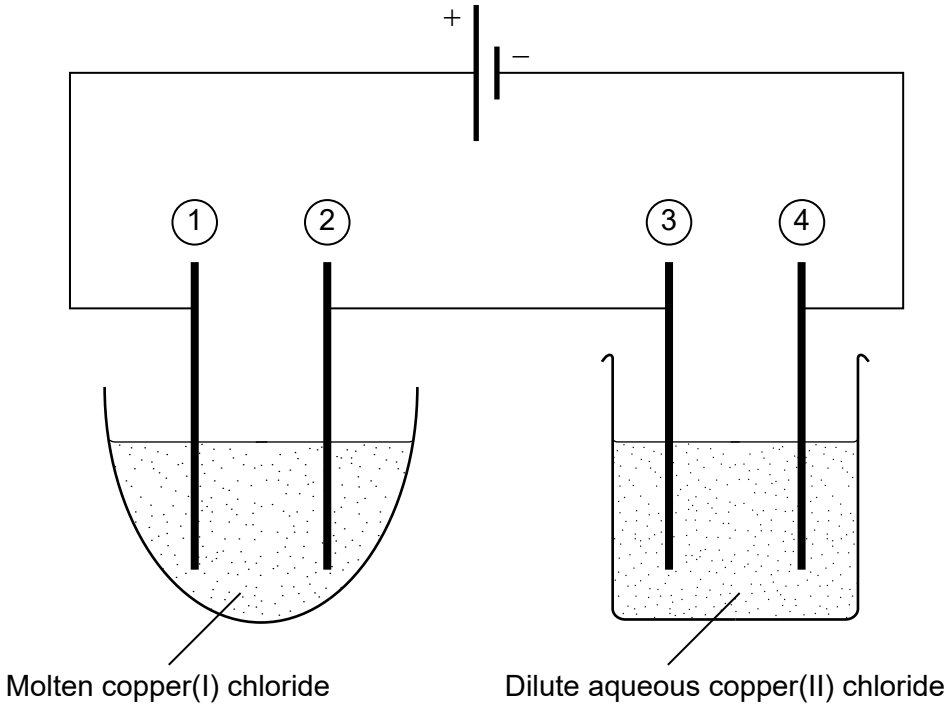
(c) Two electrolysis cells were assembled using graphite electrodes and connected in series as shown.



(i) State how current is conducted through the wires and through the electrolyte. [2]

Wires:

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Electrolyte:

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(ii) Write the half-equation for the formation of gas bubbles at electrode 1. [1]

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(iv) Deduce the half-equation for the formation of the gas identified in (c)(iii). [1]

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(e) Copper(I) chloride undergoes a disproportionation reaction, producing copper(II) chloride and copper.

$$2Cu^{+}(aq) \rightarrow Cu(s) + Cu^{2+}(aq)$$

(i) Calculate the cell potential at 298 K for the disproportionation reaction, in V, using section 24 of the data booklet. [1]

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(ii) Comment on the spontaneity of the disproportionation reaction at 298 K. [1]

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★ **7.2-S26** May 2011 TZ1 · Paper 2 · Q9 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (c), (e)(i), (e)(ii) · also 6.1, 6.2, 7.1

9. (a) The conditions used in an electrolytic cell can determine the products formed.

(i) Draw an electrolytic cell illustrating the electrolysis of molten nickel(II) bromide, $NiBr_2$. Include in the diagram the direction of the electron flow, the polarity of electrodes and state the half-equations for the product formed at each electrode. *[5]*

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(ii) Deduce the equations for the formation of the major product at the positive electrode (anode) when the following aqueous solutions are electrolysed.
- dilute sodium chloride
- concentrated sodium chloride *[2]*

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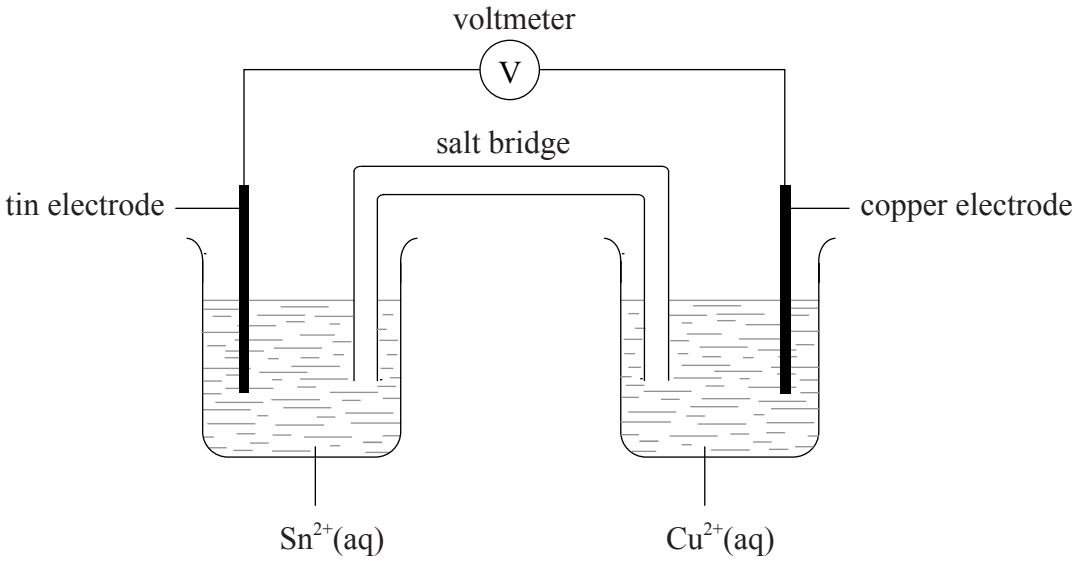
(b) A voltaic cell is constructed from two half-cells as illustrated below.



(i) Use Table 14 of the Data Booklet to deduce the equation for the spontaneous reaction occurring in this cell. *[1]*

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(ii) Calculate the standard potential for this cell. *[1]*

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(iii) State the conditions necessary for the potential of the cell to equal that calculated in part (b) (ii) using the data from Table 14. *[1]*

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(c) Using the data below and data from Table 14 of the Data Booklet, predict and explain which metal, cadmium or chromium, may be obtained by electrolysis of separate aqueous solutions of $Cd^{2+}(aq)$ ions and $Cr^{2+}(aq)$ ions. *[2]*

	$E^{\ominus}$ / V
$Cd^{2+}(aq) + 2e^{-} \rightarrow Cd(s)$	-0.40
$Cr^{2+}(aq) + 2e^{-} \rightarrow Cr(s)$	-0.91

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(e) Nitrogen monoxide may be removed from industrial emissions via a reaction with ammonia as shown by the equation below.

$$4NH_3(g) + 6NO(g) \rightarrow 5N_2(g) + 6H_2O(l)$$

(i) Deduce the oxidation number of the nitrogen in the reactants and product. *[3]*

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(ii) Deduce the oxidation and reduction half-equations and identify the oxidizing agent for the reaction. *[3]*

LESSON 7.3

Energy From Fuels

2 multiple-choice · 3 structured

Part A - Multiple choice

7.3-M3 May 2017 TZ2 · Paper 1 · Q1

1. What is the sum of the coefficients when the equation is balanced with whole numbers?

$$__C_8H_{18}(g) + __O_2(g) \rightarrow __CO(g) + __H_2O(l)$$

A. 26.5

B. 30

C. 53

D. 61

★ **7.3-M4** May 2015 TZ2 · Paper 1 · Q2

2. Which expression gives the sum of all the coefficients for the general equation for the complete combustion of hydrocarbons?

$$___ C_xH_y(g) + ___ O_2(g) \rightarrow ___ CO_2(g) + ___ H_2O(l)$$

A. $1 + x + \frac{y}{4}$

B. $1 + x + \frac{y}{2}$

C. $1 + 2x + \frac{3y}{4}$

D. $1 + 2x + \frac{3y}{2}$

Part B - Structured questions

★ **7.3-S1** May 2025 TZ3 · Paper 2 · Q9 parts (b)(i), (b)(ii), (c) · also 8.2, 9.3

9. Ethanol is a biofuel that is made from glucose. Glucose is formed by the biological fixation of carbon through photosynthesis.

(b) (i) Outline **two** advantages of using ethanol as a fuel instead of gasoline (petrol). [2]

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(ii) Calculate the enthalpy change for the combustion of ethanol in $kJ\,mol^{-1}$. Use section 12 of the data booklet. [3]

$$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$$

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(c) Calculate the increase in temperature in a sample of 500.0 cm^3 of water heated by the combustion of 4.00 g of ethanol. Use sections 2 and 14 of the data booklet. [2]

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★ **7.3-S3** May 2024 TZ1 · Paper 3 · Q13 parts (a), (b), (c), (f)

13. Methane clathrate looks like ice but burns when ignited because it turns back to water and methane.

(a) Write an equation for the complete combustion of methane. [1]

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(b) The formula of methane clathrate is $(CH_4)_4(H_2O)_{23}$ and its molar mass is 478.7 $g\,mol^{-1}$. Calculate the specific energy of methane clathrate. Use sections 1, 6 and 13 of the data booklet. [2]

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(c) Suggest why methane clathrate has a lower specific energy but higher energy density than methane. [2]

Lower specific energy:

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Higher energy density:

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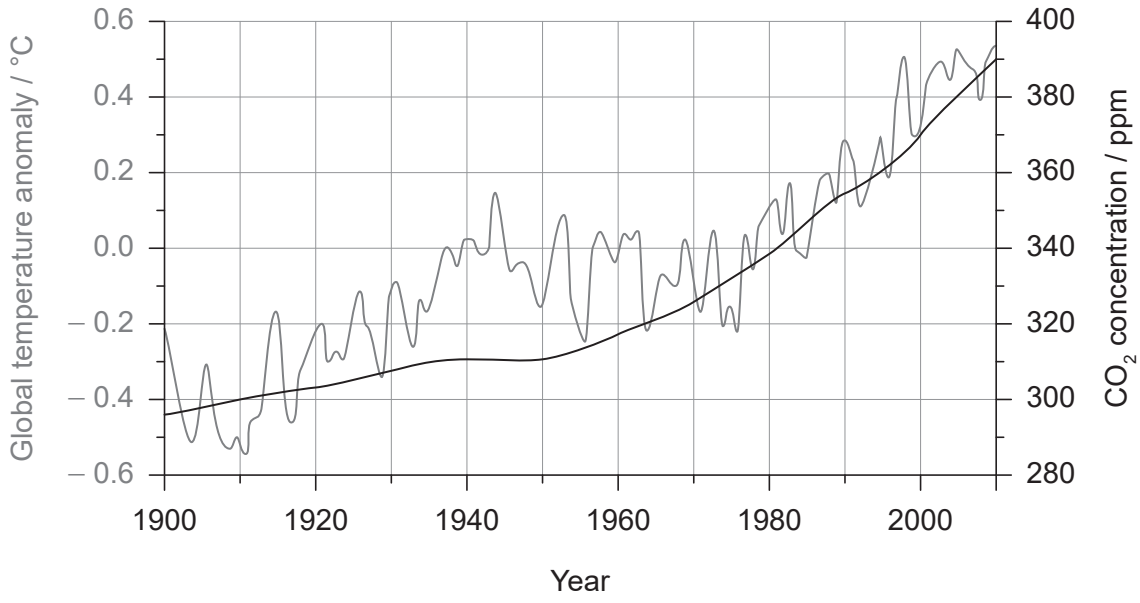
(f) The graph shows the increase in temperature and the level of carbon dioxide in the atmosphere.



Discuss how this data could be used to both support and refute the theory that carbon dioxide is a cause of global warming. [2]

Support:

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Refute:

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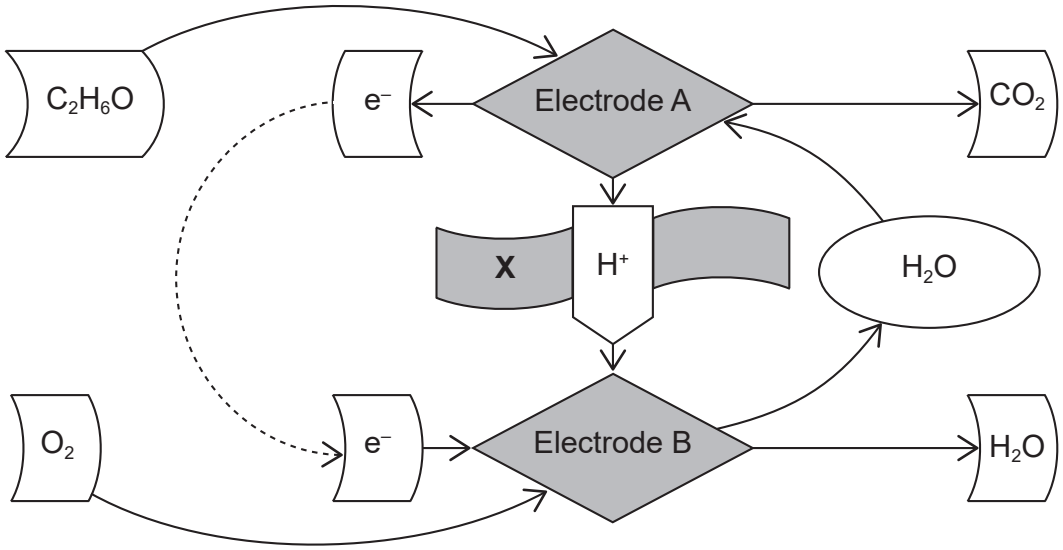
7.3-S19 May 2019 TZ1 · Paper 3 · Q17 parts (b)(i), (b)(ii), (b)(iii), (c)

17. E10 is composed of 10 % ethanol and 90 % normal unleaded fuel.

(b) (i) Ethanol can be used in a direct-ethanol fuel cell (DEFC) as illustrated by the flow chart.



[Source: https://commons.wikimedia.org/wiki/File:Direct_Ethanol_Fuel_Cell.svg, by Fordi]

Deduce the half-equations occurring at electrodes A and B. [2]

Electrode A:

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Electrode B:

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(c) Biodiesel containing ethanol can be made from renewable resources.

Suggest **one** environmental disadvantage of producing biodiesel from renewable resources. [1]

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LESSON 8.1

Thermochemical Equations

3 multiple-choice · 1 structured

Part A - Multiple choice

8.1-M4 Nov 2023 TZ1 · Paper 1 · Q14

14. Which reactions release heat?

I. $C(s) + O_2(g) \rightarrow CO_2(g)$

II. $Na^+(g) + e^- \rightarrow Na(g)$

III. $NH_3(g) \rightarrow NH_3(l)$

A. I and II only

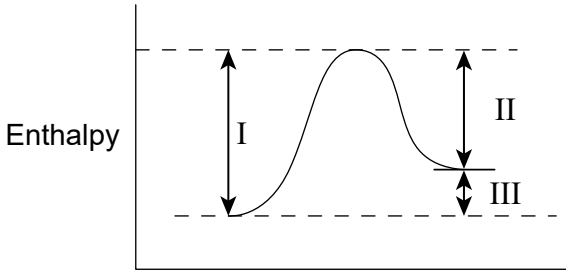
B. I and III only

C. II and III only

D. I, II and III

8.1-M5 Nov 2023 TZ1 · Paper 1 · Q16

16. Which expression represents the calculation of ΔH?



A. I–II

B. II–I

C. I–III

D. II–III

★ **8.1-M9** Nov 2020 · Paper 1 · Q15

15. Which statements about bond strength and activation energy are correct for this reaction?

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) \qquad \Delta H^{\ominus} = -891\,kJ$$

	Relative bond strength	Relative magnitude of activation energy, E_a
A.	products < reactants	forward > reverse
B.	products > reactants	forward < reverse
C.	products > reactants	forward > reverse
D.	products < reactants	forward < reverse

Part B - Structured questions

★ **8.1-S8** May 2012 TZ2 · Paper 2 · Q8 parts (a)(i), (a)(ii), (a)(iii), (c) · also 8.2

8. (a) The equation for the reaction between sodium hydroxide, NaOH, and nitric acid, HNO_3, is shown below.

$$NaOH(aq) + HNO_3(aq) \rightarrow NaNO_3(aq) + H_2O(l) \qquad \Delta H = -57.6\ kJ\ mol^{-1}$$

(i) Sketch and label an enthalpy level diagram for this reaction. *[3]*

(ii) Deduce whether the reactants or the products are more energetically stable, stating your reasoning. *[1]*

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(iii) Calculate the change in heat energy, in kJ, when 50.0 cm^3 of 2.50 $mol\,dm^{-3}$ sodium hydroxide solution is added to excess nitric acid. *[2]*

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(c) When 5.35 g ammonium chloride, $NH_4Cl(s)$, is added to 100.0 cm^3 of water, the temperature of the water decreases from 19.30 °C to 15.80 °C. Determine the enthalpy change, in kJ mol^{-1}, for the dissolving of ammonium chloride in water. *[3]*

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LESSON 8.2

Calorimetry

4 multiple-choice · 2 structured

Part A - Multiple choice

★ **8.2-M1** May 2025 TZ1 · Paper 1A · Q23

23. 0.050 mol of sodium hydrogencarbonate was added to 50 cm^3 of 2.0 mol dm^{-3} solution of citric acid.

$$C_6H_8O_7(aq) + 3NaHCO_3(s) \rightarrow C_6H_5O_7Na_3(aq) + 3CO_2(g) + 3H_2O(l)$$

The initial temperature of the solution was 298 K and the final temperature was 297 K.
What is the enthalpy of this reaction in kJ mol^{-1}? ($c_w = 4.18$ J g^{-1} K^{-1}, $Q = mc\Delta T$)

A. +4.18

B. –4.18

C. +12.5

D. –12.5

8.2-M8 May 2023 TZ2 · Paper 1 · Q15

15. When 100 cm^3 of 1.0 mol dm^{-3} HCl is mixed with 100 cm^3 of 1.0 mol dm^{-3} NaOH, the temperature of the resulting solution increases by 5.0 °C. What will be the temperature change, in °C, when 50 cm^3 of 2.0 mol dm^{-3} HCl is mixed with 50 cm^{-3} of 2.0 mol dm^{-3} NaOH?

A. 2.5

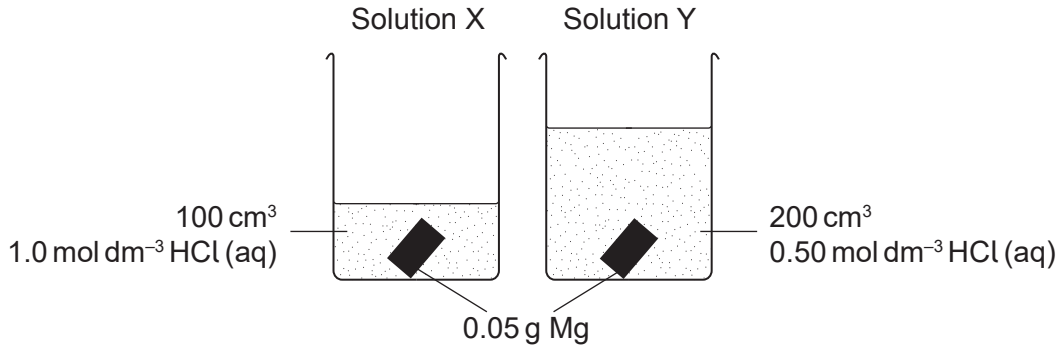
B. 5.0

C. 10

D. 20

8.2-M10 May 2022 TZ2 · Paper 1 · Q13

13. Which statement is correct about identical pieces of magnesium added to two solutions, X and Y, containing hydrochloric acid at the same temperature?



A. Solution X will reach a higher maximum temperature.

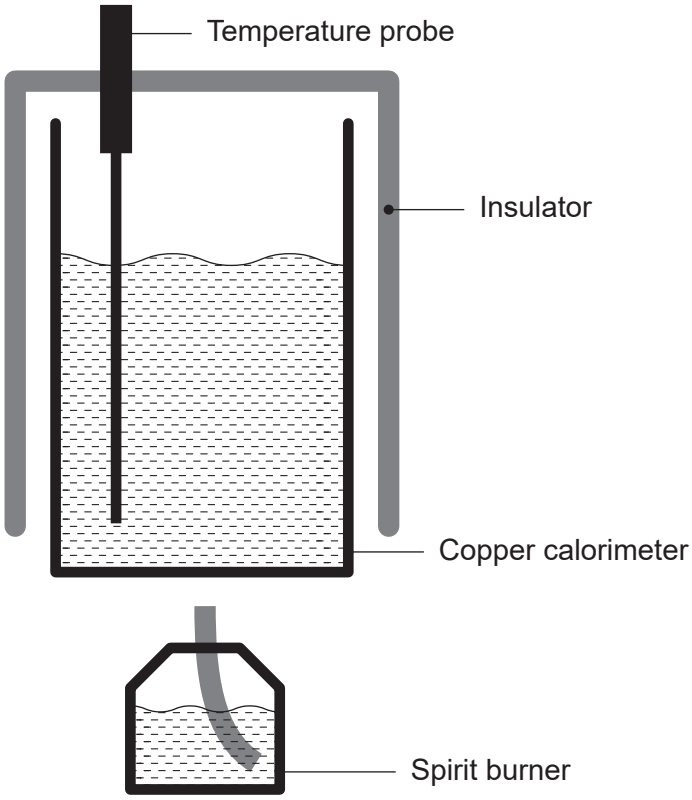
B. Solution Y will reach a higher maximum temperature.

C. Solutions X and Y will have the same temperature rise.

D. It is not possible to predict whether X or Y will have the higher maximum temperature because we cannot identify the limiting reactant.

8.2-M12 May 2021 TZ1 · Paper 1 · Q38

38. The enthalpy of combustion of a fuel was determined using the calorimeter shown. The final result was lower than the literature value.



Which factors could have contributed to this error?

I. Not all heat from the combustion was transferred to the calorimeter.

II. Incomplete combustion occurred.

III. The temperature probe touched the bottom of the calorimeter.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

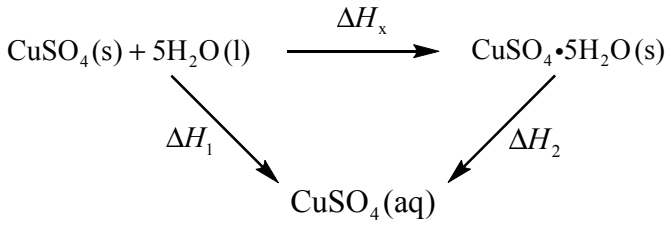
Part B - Structured questions

8.2-S17 May 2011 TZ2 · Paper 2 · Q1 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i), (b)(ii), (b)(iii), (c)(i), (c)(ii) · also 1.2, 9.1

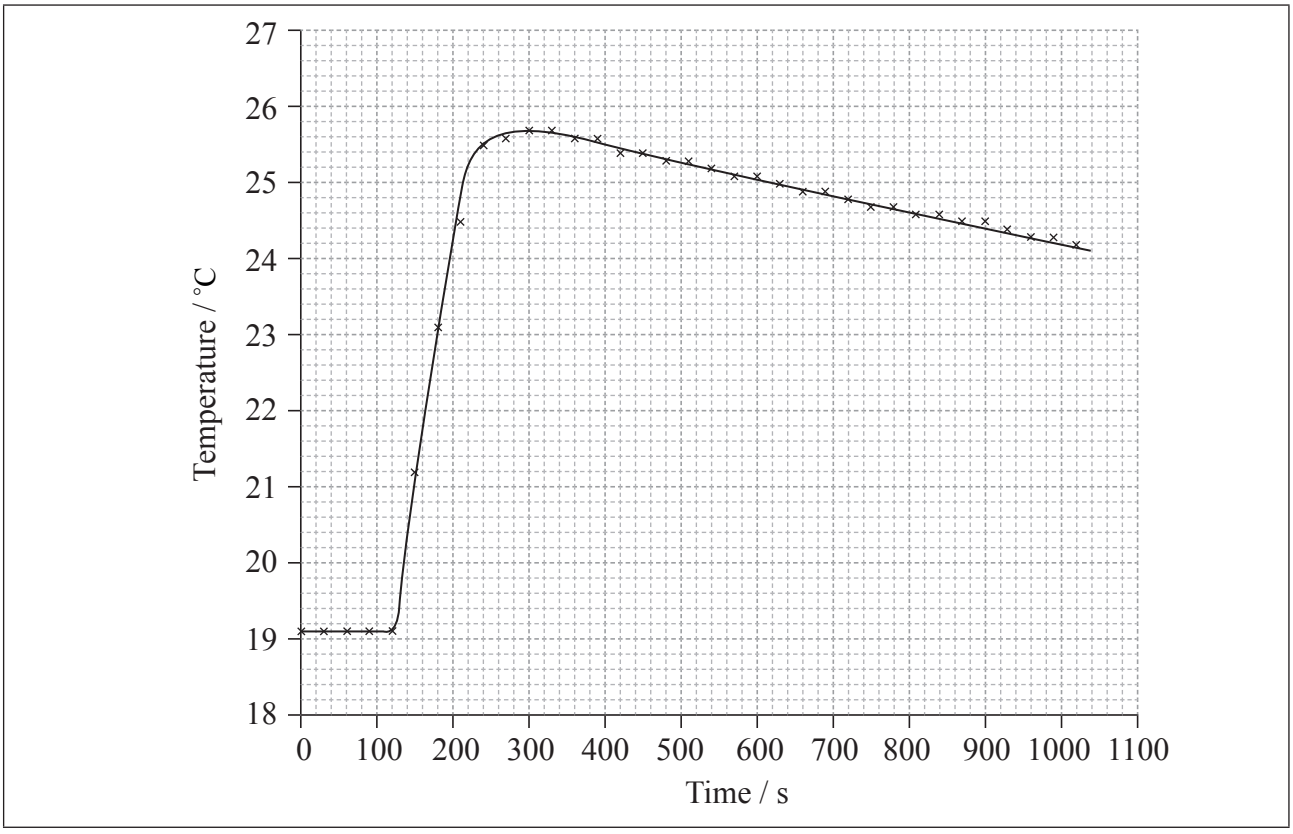
1. If white anhydrous copper(II) sulfate powder is left in the atmosphere it slowly absorbs water vapour giving the blue pentahydrated solid.

$$\underset{\text{(anhydrous)}}{CuSO_4(s)} + 5H_2O(l) \rightarrow \underset{\text{(pentahydrated)}}{CuSO_4 \cdot 5H_2O(s)}$$

It is difficult to measure the enthalpy change for this reaction directly. However, it is possible to measure the heat changes directly when both anhydrous and pentahydrated copper(II) sulfate are separately dissolved in water, and then use an energy cycle to determine the required enthalpy change value, ΔH_x, indirectly.



(a) To determine ΔH_1 a student placed 50.0 g of water in a cup made of expanded polystyrene and used a data logger to measure the temperature. After two minutes she dissolved 3.99 g of anhydrous copper(II) sulfate in the water and continued to record the temperature while continuously stirring. She obtained the following results.



(iii) Calculate the heat change, in kJ, when 3.99 g of anhydrous copper(II) sulfate is dissolved in the water. *[2]*

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(iv) Determine the value of ΔH_1 in kJ mol^{-1}. *[1]*

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(b) To determine ΔH_2, 6.24 g of pentahydrated copper(II) sulfate was dissolved in 47.75 g of water. It was observed that the temperature of the solution decreased by 1.10 °C.

(i) Calculate the amount, in mol, of water in 6.24 g of pentahydrated copper(II) sulfate. *[2]*

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(ii) Determine the value of ΔH_2 in kJ mol^{-1}. *[2]*

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(iii) Using the values obtained for ΔH_1 in (a) (iv) and ΔH_2 in (b) (ii), determine the value for ΔH_x in kJ mol^{-1}. *[1]*

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(c) The magnitude (the value without the + or – sign) found in a data book for ΔH_x is 78.0 kJ mol^{-1}.

(i) Calculate the percentage error obtained in this experiment. (If you did not obtain an answer for the experimental value of ΔH_x then use the value 70.0 kJ mol^{-1}, but this is **not** the true value.) *[1]*

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(ii) The student recorded in her qualitative data that the anhydrous copper(II) sulfate she used was pale blue rather than completely white. Suggest a reason why it might have had this pale blue colour and deduce how this would have affected the value she obtained for ΔH_x. *[2]*

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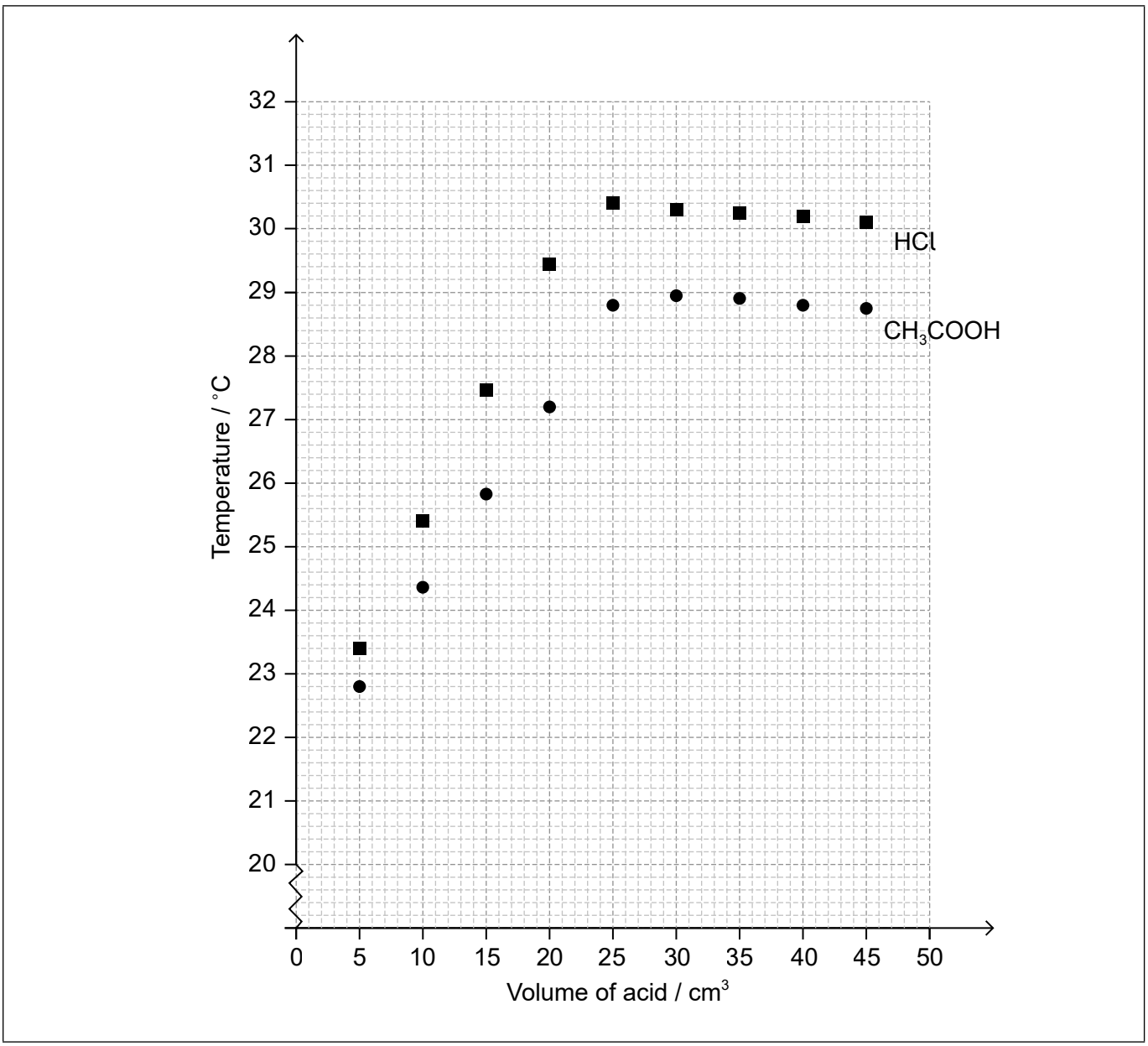
8.2-S8 Nov 2017 · Paper 2 · Q1 parts (a), (b), (c), (d)(i), (d)(ii) · also 1.2

1. A student titrated two acids, hydrochloric acid, HCl (aq) and ethanoic acid, CH_3COOH (aq), against 50.0 cm^3 of 0.995 mol dm^{-3} sodium hydroxide, NaOH (aq), to determine their concentration. The temperature of the reaction mixture was measured after each acid addition and plotted against the volume of each acid.



(a) Using the graph, estimate the initial temperature of the solutions. [1]

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(b) Determine the maximum temperature reached in each experiment by analysing the graph. [2]

HCl:

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CH_3COOH:

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(c) Calculate the concentration of ethanoic acid, CH_3COOH, in $mol\,dm^{-3}$. [2]

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(d) (i) Determine the heat change, q, in kJ, for the neutralization reaction between ethanoic acid and sodium hydroxide.
Assume the specific heat capacities of the solutions and their densities are those of water. [2]

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(ii) Calculate the enthalpy change, ΔH, in $kJ\,mol^{-1}$, for the reaction between ethanoic acid and sodium hydroxide. [2]

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LESSON 9.1

Hess's Law

3 multiple-choice · 1 structured

Part A - Multiple choice

9.1-M1 Nov 2024 · Paper 1 · Q16

16. Consider the following equations:

$$H_2(g) + F_2(g) \rightarrow 2HF(g) \qquad \Delta H^{\ominus} = -x\,kJ$$

$$C(s) + 2F_2(g) \rightarrow CF_4(g) \qquad \Delta H^{\ominus} = -y\,kJ$$

$$2C(s) + 2H_2(g) \rightarrow C_2H_4(g) \qquad \Delta H^{\ominus} = +z\,kJ$$

What is the $\Delta H^{\ominus}$ of the reaction?

$$C_2H_4(g) + 6F_2(g) \rightarrow 2CF_4(g) + 4HF(g)$$

A. $-x - y - z$

B. $-x - y + z$

C. $-2x + 2y - z$

D. $-2x - 2y - z$

★ **9.1-M2** Nov 2023 TZ1 · Paper 1 · Q15

15. Which expression represents the calculation used to obtain the $\Delta H^{\ominus}$ value for the conversion of oxygen to one mole of ozone (O_3)?

		$\Delta H^{\ominus}$, kJ
Eqn (i)	$2CO_2 \rightarrow 2CO + O_2$	+566
Eqn (ii)	$3CO + O_3 \rightarrow 3CO_2$	–992

A. $-566 - 992$

B. $-566 + 992$

C. $1.5 \times (-566) + 992$

D. $1.5 \times (-566) - 992$

9.1-M6 May 2018 TZ1 · Paper 1 · Q14

14. What is the enthalpy of combustion of butane in **kJ mol^{-1}**?

$$2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(l)$$

$$C(s) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H = x \text{ kJ}$$

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l) \qquad \Delta H = y \text{ kJ}$$

$$4C(s) + 5H_2(g) \rightarrow C_4H_{10}(g) \qquad \Delta H = z \text{ kJ}$$

A. $4x + 5y - z$

B. $4x + 5y + z$

C. $8x + 10y - 2z$

D. $8x + 5y + 2z$

Part B - Structured questions

9.1-S2 May 2017 TZ2 · Paper 2 · Q9 parts (a)(i), (a)(ii) · also 8.2

9. The Bombardier beetle sprays a mixture of hydroquinone and hydrogen peroxide to fight off predators. The reaction equation to produce the spray can be written as:

$$\underset{\text{hydroquinone}}{C_6H_4(OH)_2(aq)} + H_2O_2(aq) \rightarrow \underset{\text{quinone}}{C_6H_4O_2(aq)} + 2H_2O(l)$$

(a) (i) Calculate the enthalpy change, in kJ, for the spray reaction, using the data below. [2]

$$C_6H_4(OH)_2(aq) \rightarrow C_6H_4O_2(aq) + H_2(g) \qquad \Delta H^{\ominus} = +177.0\,\text{kJ}$$

$$2H_2O(l) + O_2(g) \rightarrow 2H_2O_2(aq) \qquad \Delta H^{\ominus} = +189.2\,\text{kJ}$$

$$H_2O(l) \rightarrow H_2(g) + \frac{1}{2}O_2(g) \qquad \Delta H^{\ominus} = +285.5\,\text{kJ}$$

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(ii) The energy released by the reaction of one mole of hydrogen peroxide with hydroquinone is used to heat 850 cm^3 of water initially at 21.8°C. Determine the highest temperature reached by the water.

Specific heat capacity of water = 4.18 kJ kg^{-1} K^{-1}.

(If you did not obtain an answer to part (i), use a value of 200.0 kJ for the energy released, although this is not the correct answer.) [2]

LESSON 9.2

Enthalpies of Formation & Combustion

2 multiple-choice · 1 structured

Part A - Multiple choice

★ **9.2-M11** Nov 2017 · Paper 1 · Q16

16. What is the standard enthalpy of formation, in kJ mol^{-1}, of IF (g)?

$$\text{IF}_7\text{(g)} + \text{I}_2\text{(s)} \rightarrow \text{IF}_5\text{(g)} + 2\text{IF (g)} \qquad \Delta H^{\ominus} = -89\,\text{kJ}$$

$\Delta H_f^{\ominus}(\text{IF}_7) = -941\,\text{kJ mol}^{-1}$
$\Delta H_f^{\ominus}(\text{IF}_5) = -840\,\text{kJ mol}^{-1}$

A. -190

B. -95

C. $+6$

D. $+95$

9.2-M13 May 2016 · Paper 1 · Q16

16. The equation for the formation of ethyne is:

$$2\text{C (s)} + \text{H}_2\text{(g)} \rightarrow \text{C}_2\text{H}_2\text{(g)}$$

What is the enthalpy change, in kJ, for this reaction using the enthalpy of combustion data below?

Reaction	$\Delta H^{\ominus}$ / kJ
$\text{C (s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$	-394
$2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{H}_2\text{O (l)}$	-572
$2\text{C}_2\text{H}_2\text{(g)} + 5\text{O}_2\text{(g)} \rightarrow 4\text{CO}_2\text{(g)} + 2\text{H}_2\text{O (l)}$	-2602

A. $2 \times (-394) + \frac{1}{2}(-572) - \frac{1}{2}(-2602)$

B. $2 \times (-394) + (-572) - (-2602)$

C. $2 \times (-394) + \frac{1}{2}(-572) + \frac{1}{2}(-2602)$

D. $2 \times (-394) + (-572) + (-2602)$

Part B - Structured questions

9.2-S2 Nov 2023 TZ1 · Paper 2 · Q4 parts (a)(i), (b) · also 3.1

4. Carbon disulfide, CS_2, undergoes gas phase hydrolysis according to the overall equation

$$CS_2(g) + 2H_2O(g) \rightleftharpoons CO_2(g) + 2H_2S(g)$$

(a) (i) Calculate the enthalpy change in this reaction from section 12 of the data booklet and the given values: [2]

	$CS_2(g)$	$H_2S(g)$
$\Delta H_f^\ominus$	$+88.7\,kJ\,mol^{-1}$	$-20.6\,kJ\,mol^{-1}$

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(b) Deduce the molecular geometries of CS_2 and H_2S, and the reason why they are different. [2]

Molecular geometry CS_2: . . . . . .

Molecular geometry H_2S: . . . . . .

Reason for difference: . . . . . .

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LESSON 9.3

Bond Enthalpies

2 multiple-choice · 3 structured

Part A - Multiple choice

★ **9.3-M5** May 2016 · Paper 1 · Q17

17. Which equation represents the average bond enthalpy of the Si–H bond in SiH_4?

A. $SiH_4(g) \rightarrow SiH_3(g) + H(g)$

B. $\frac{1}{4}SiH_4(g) \rightarrow \frac{1}{4}Si(g) + H(g)$

C. $SiH_4(g) \rightarrow SiH_3(g) + \frac{1}{2}H_2(g)$

D. $SiH_4(g) \rightarrow Si(g) + 4H(g)$

9.3-M6 May 2015 TZ2 · Paper 1 · Q15

15. Which enthalpy changes can be calculated using **only** bond enthalpy data?

I. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

II. $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$

III. $CH_4(g) + Cl_2(g) \rightarrow CH_3Cl(g) + HCl(g)$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

Part B - Structured questions

9.3-S1 May 2025 TZ1 · Paper 2 · Q8 parts (c)(v), (f)(i), (f)(ii), (f)(iii) · also 5.2, 9.2

8. Organic compounds have many industrial applications.

(c) An alkene such as ethene can be used as starting material for a range of compounds.

(v) Explain, in terms of the intermolecular forces present, the trend in the boiling points of the first four alkenes. [2]

Alkene	Boiling point / K
ethene	169
propene	225
but-1-ene	267
pent-1-ene	303

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(f) Ethene reacts with steam to produce ethanol.

$$C_2H_4(g) + H_2O(g) \rightarrow C_2H_5OH(g)$$

(i) Calculate the enthalpy, in kJ, of the reaction using section 12 of the data booklet. [3]

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(ii) Calculate the enthalpy of the reaction, in kJ. Use section 13 of the data booklet and $\Delta H_f^{\ominus}$ of $CH_3CH_2OH(g) = -235\,kJ\,mol^{-1}$. [2]

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(iii) Outline why the enthalpies calculated in (i) and (ii) are different. [1]

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9.3-S8 May 2018 TZ1 · Paper 2 · Q3 parts (c)(i), (c)(ii), (c)(iii), (d) · also 4.1, 9.2

3. This question is about ethene, C_2H_4, and ethyne, C_2H_2.

(c) (i) Under certain conditions, ethyne can be converted to benzene.

Determine the standard enthalpy change, $\Delta H^{\ominus}$, for the reaction stated, using section 11 of the data booklet. [2]

$$3C_2H_2(g) \rightarrow C_6H_6(g)$$

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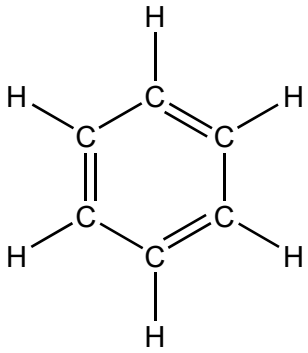

State one piece of physical evidence that this structure is **incorrect**. [1]

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★ **9.3-S13** Nov 2012 · Paper 2 · Q5 parts (a), (b)(i), (b)(ii), (b)(iii), (b)(iv), (b)(v), (c)(ii), (c)(iii) · also 3.1, 4.1, 4.2, 9.2

5. The strength of a covalent bond is measured in terms of its bond enthalpy.

(a) Define the term *average bond enthalpy*. *[2]*

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(i) Calculate the enthalpy change when 10.0 g of the compound decomposes, using average bond enthalpy data from Table 10 of the Data Booklet and the following additional average bond enthalpy data at 298 K. *[4]*

Bond	**ΔH / kJ mol^{-1}**
C≡O	1072
N–O	201
N=O	607

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(iv) Determine the number of σ and π bonds in 1,3,5,7-tetranitro-1,3,5,7-tetrazocane, using the Lewis structure shown on page 16. *[2]*

σ bonds: . . . . . .

π bonds: . . . . . .

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(iii) The standard enthalpy change of formation of CO (g) is –111 kJ mol^{-1}. Using Table 11 of the Data Booklet, determine the enthalpy change of the reaction, in kJ mol^{-1}. *[1]*

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LESSON 9.4

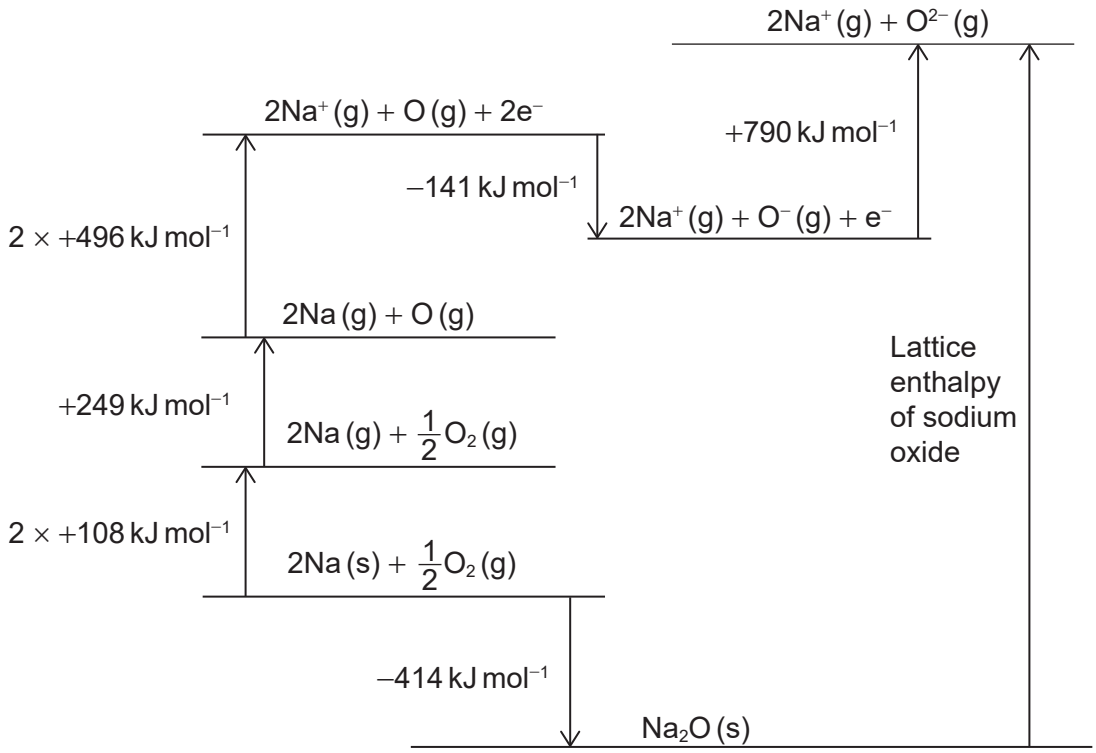
Born-Haber Cycles

2 multiple-choice · 2 structured

Part A - Multiple choice

9.4-M6 Nov 2021 · Paper 1 · Q16

16. Consider the Born–Haber cycle for the formation of sodium oxide:



What is the lattice enthalpy, in $kJ\,mol^{-1}$, of sodium oxide?

A. $414 + 2(108) + 249 + 2(496) - 141 + 790$

B. $414 + 2(108) + 249 + 2(496) + 141 + 790$

C. $-414 + 2(108) + 249 + 2(496) - 141 + 790$

D. $-414 - 2(108) - 249 - 2(496) + 141 - 790$

9.4-M9 May 2019 TZ1 · Paper 1 · Q17

17. Which equation represents the standard enthalpy of atomization of bromine, Br_2?

A. $\frac{1}{2}Br_2(l) \rightarrow Br(g)$

B. $Br_2(l) \rightarrow 2Br(g)$

C. $Br_2(l) \rightarrow 2Br(l)$

D. $\frac{1}{2}Br_2(l) \rightarrow Br(l)$

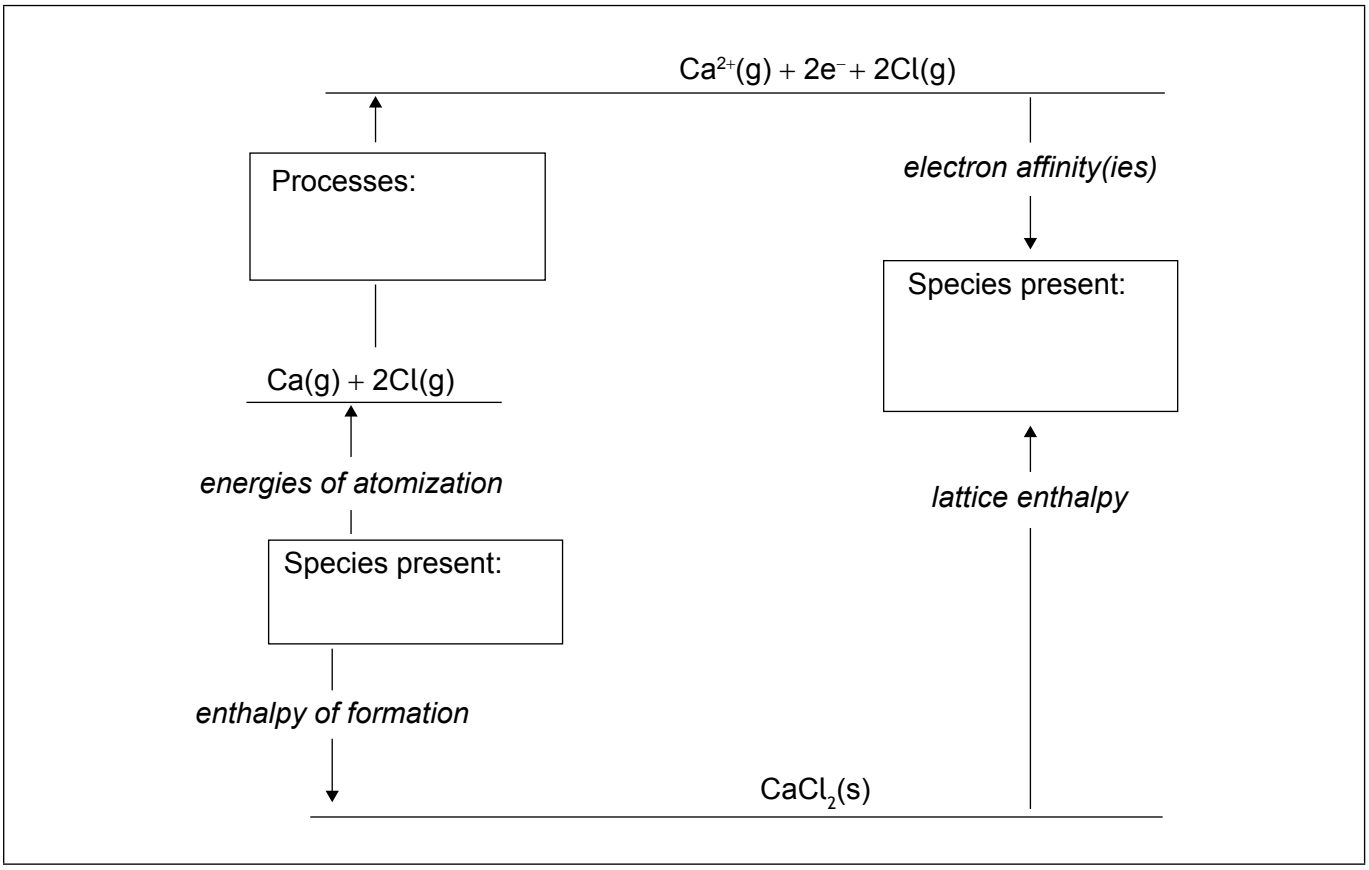
Part B - Structured questions

9.4-S1 May 2025 TZ1 · Paper 2 · Q4 parts (c)(i), (c)(ii)

4. 3.162 g of calcium carbonate, $CaCO_3(s)$, is reacted with 20.0 cm^3 of 4.00 mol dm^{-3} hydrochloric acid, HCl(aq).

(c) Calcium carbonate and calcium chloride are ionic compounds.

(i) Annotate the Born–Haber cycle for calcium chloride, $CaCl_2$, by filling the names of processes and formulas of species, including state symbols, in the boxes. [3]



(ii) Outline why the lattice enthalpy of calcium chloride is lower than the lattice enthalpy of magnesium chloride. [1]

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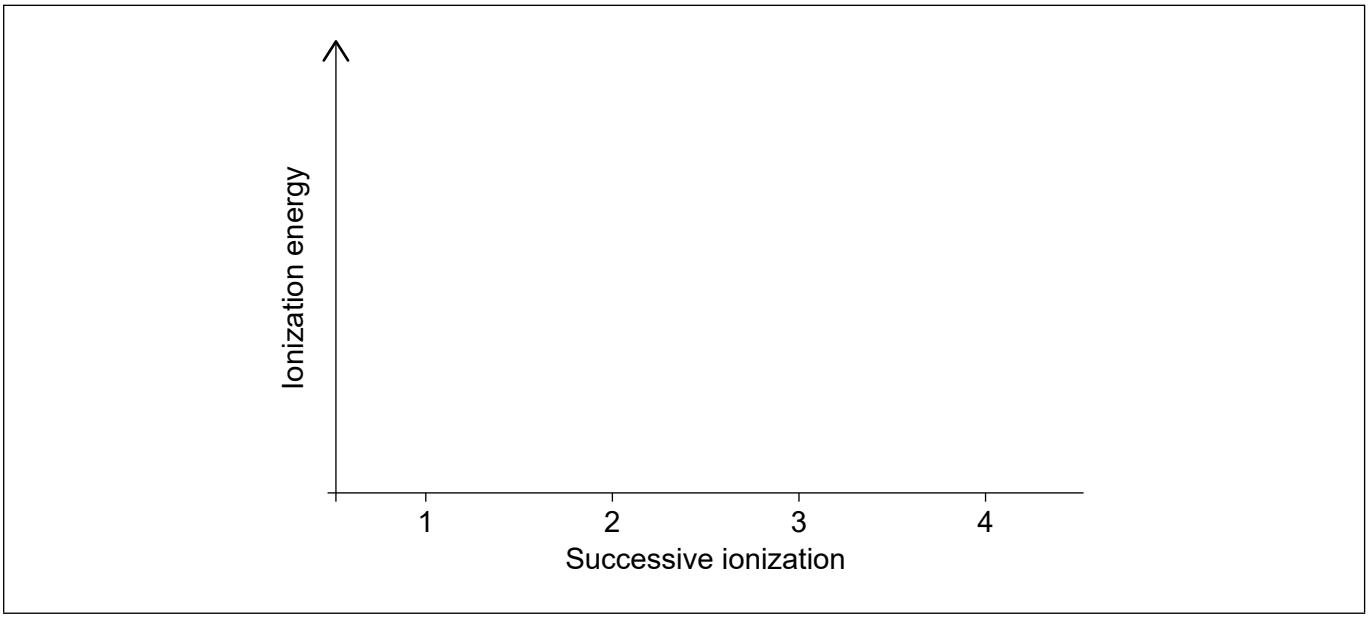
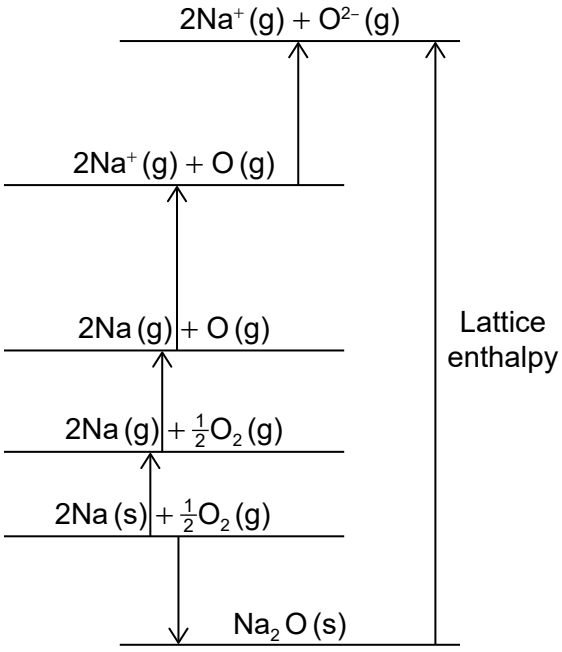
9.4-S3 May 2019 TZ1 · Paper 2 · Q3 parts (a), (d)(i), (d)(ii), (d)(iii), (f), (g)(i), (g)(ii), (h), (k) · also 1.2, 2.1, 4.1, 6.1, 9.2, 9.3

3. This question is about sodium and its compounds.

(a) Plot the relative values of the first four ionization energies of sodium. [1]



(d) The Born-Haber cycle for sodium oxide is shown (not to scale).



(i) Calculate values for the following changes using section 8 of the data booklet. [2]

$$\Delta H_{\text{atomisation}}\,(\text{Na}) = 107\,\text{kJ mol}^{-1}$$

$$\Delta H_{\text{atomisation}}\,(\text{O}) = 249\,\text{kJ mol}^{-1}$$

$\frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{O}^{2-}(\text{g})$:

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$\text{Na}(\text{s}) \rightarrow \text{Na}^{+}(\text{g})$:

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(ii) The standard enthalpy of formation of sodium oxide is $-414\,\text{kJ mol}^{-1}$. Determine the lattice enthalpy of sodium oxide, in kJ mol^{-1}, using section 8 of the data booklet and your answers to (d)(i). [2]

(If you did not get answers to (d)(i), use $+850\,\text{kJ mol}^{-1}$ and $+600\,\text{kJ mol}^{-1}$ respectively, but these are not the correct answers.)

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(iii) Justify why K_2O has a lower lattice enthalpy (absolute value) than Na_2O. [1]

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(f) Sodium peroxide, Na_2O_2, is formed by the reaction of sodium oxide with oxygen.

$$2Na_2O(s) + O_2(g) \rightarrow 2Na_2O_2(s)$$

Calculate the percentage yield of sodium peroxide if 5.00 g of sodium oxide produces 5.50 g of sodium peroxide. [2]

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(g) Sodium peroxide is used in diving apparatus to produce oxygen from carbon dioxide.

$$2Na_2O_2(s) + 2CO_2(g) \rightarrow 2Na_2CO_3(s) + O_2(g)$$

(i) Determine the enthalpy change, ΔH, in kJ, for this reaction using data from the table and section 12 of the data booklet. [3]

	ΔH_f / kJ mol^{-1}
$Na_2O_2(s)$	–510.9
$Na_2CO_3(s)$	–1130.7

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(k) State the oxidation number of carbon in sodium carbonate, Na_2CO_3. [1]

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